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PortfolioSemester

Aqua Green

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Egypt's Grand Challenges represent the most significant problems that shape the country's future stability, growth, and security. These challenges include: Improve the use of alternative energies, Recycle garbage and waste for economic and environmental purposes, Deal with urban congestion and its consequences, Work to eradicate public health issues/disease, Increase the industrial and agricultural bases of Egypt, Address and reduce pollution fouling our air, water and soil, Improve uses of arid areas, Manage and increase the sources of clean water, Deal with population growth and its consequences, Improve the scientific and technological environment for all, and Reduce and adapt to the effect of climate change, **As shown in Figure 1**.

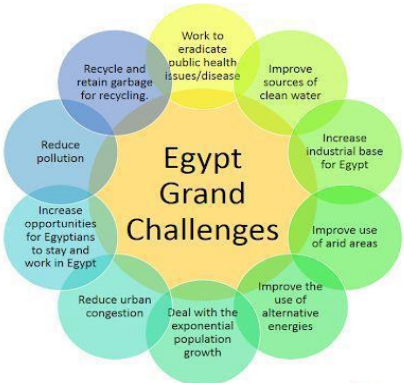


Figure 1: Egypt Grand Challenges

Together, these eleven issues create a massive set of pressures that effects on every sector, especially agriculture, industry, and public services. They stress natural resources, reduce productivity, decrease opportunities for innovation, and slow sustainable development needed for a growing population.

These challenges are critical to address because they directly affect Egypt's ability to build a modern, productive, and sustainable future. Improving the scientific and technological environment for all is essential for creating skilled workers, supporting innovation, and raising the country's global competitiveness. At the same time, addressing and reducing the pollution that harms Egypt's air, water, and soil is necessary for protecting public health, maintaining natural resources, and lowering economic losses caused by environmental damage. Egypt should also reduce and adapt to the effects of climate change, as rising temperatures, shifting weather patterns, and increasing droughts endanger agriculture, infrastructure, and national stability. Finally, recycling garbage and waste for economic and environmental purposes can transform a major national load into an opportunity that saves resources, creates jobs, and supports cleaner, healthier cities. Solving these four connected challenges is essential for establishing a strong scientific base, protecting the environment, and ensuring a stable path of development for Egypt.

Chapter One: Present and Justify a Problem and Solution Requirements

Egypt Grand Challenge

Population growth and its consequences

Population growth is increasing in the number of people in the country. it isn't measured by birth rate only, but it has another factor like migration and immigration, and death rate to give it a formula, population growth = (Birth rate +Migration) - (Death rate +Immigration), and it has many causes like high birth rate the main factor of the increasing and increasing in good care health that reduce death rate that make population growth in continuous increasing without decreasing, and the migration that increase the population, which increases the population growth every year. It also has many impacts, like racial conflicts that affect the security in Egypt and on the Egyptian economic that make a heavy load on the government. **As Shown in Figure 2.**

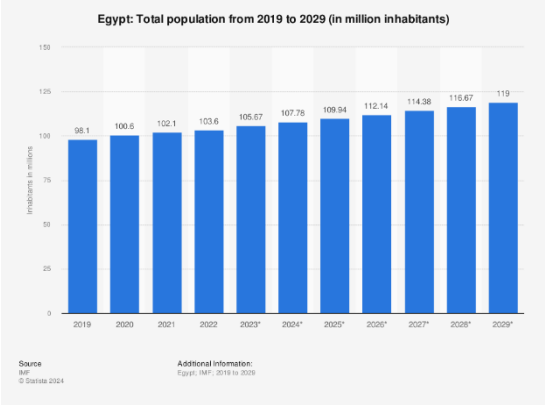


Figure 2 shows the tatal population ***Statista. (2024b, July 4).***

The population growth was 107.78 in 2024. If compared to the previous years, it will be the biggest one ever. This explains why there are many troubles in Egypt in economic terms and in the social aspect, so the government must enact laws to prevent this hard trouble.

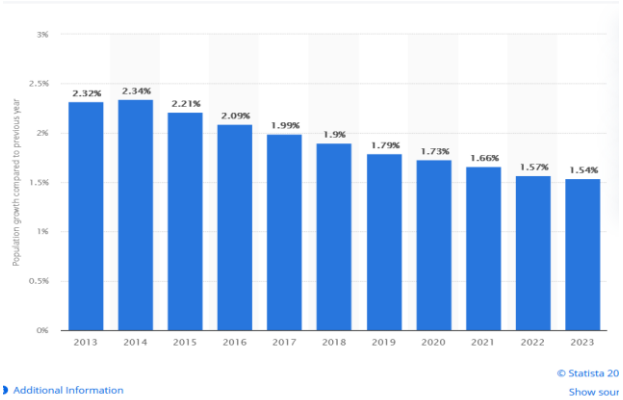


Figure 3 shows the population growth percentage, Statista. (2024b, September 27).

As shown in Figure 3, population growth increased in 2023 by 1.54% compared to the previous that mean 1.54% from total of the total population growth in 2022, as the total in 2022 was 105.67 million multiply 1.54% that mean increased over 1 million compared to the previous year, and that was the biggest increase.

CAUSES:

High Birth Rates in Egypt**:**

Because there are places where there is illiteracy and ignorance, it has a **high birth rate,** with children under 18 years equaling 40.9 million in 2021, and the number of children in the countryside is equal to 25.3 million, **As Shown in Figure 4**

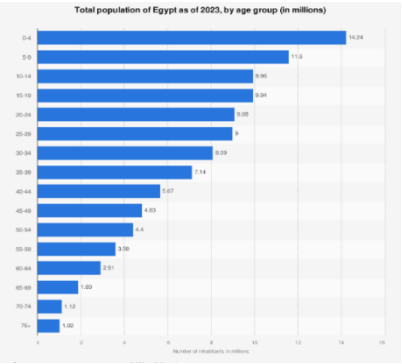


Figure 4 shows the total population of Egypt in 2023

The children from newborn to 19 have a population equal 45.74% on the other hand, the adults from 20 to 75+ have a population equal 54.26% in 2023 so that mean the birth rate increased significantly in this year, as it almost to reach half the population in Egypt

Good health care in Egypt

There is good health care in Egypt, as there were 91,454 professional physicians in the governmental healthcare sector in 2020. Furthermore, this number saw a gradual increase over the years, reaching an overall increase of 18.38 percent. know with just over 103,000 professional physicians.

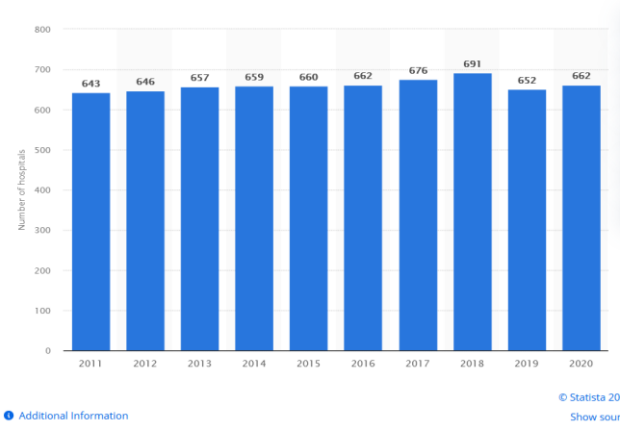


Figure 5 The Number of hospitals in Egypt's public and private healthcare sector in 2020

As shown in figure 5 The Number of hospitals in Egypt's public and private healthcare sector in 2020

Reached 662 hospitals in Egypt, and it is continuously increasing.

Migration

Due to there being many peoples migrate to Egypt to find better job **as shown in Figure 6,** there are 544 thousand immigrants in 2020, with over 500,000 people from Sudan migrating to Egypt, 962.432 from Saudi Arabia, and 899.612 from the US. From April 2023 to March 2024, around 1.5 million Syrians were in Egypt, most of whom arrived after the 2011 revolution according to UNCHR

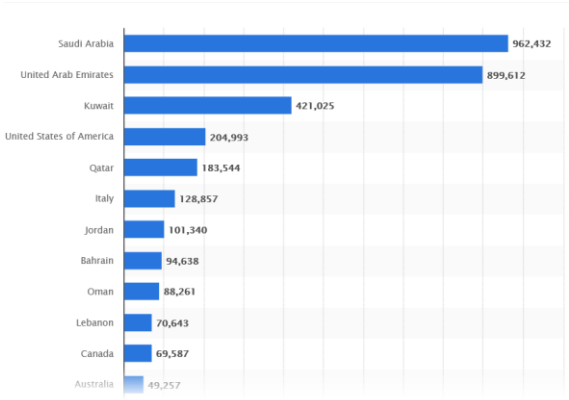


Figure 6 shows Migration, source statista

Impacts

Effect on the Egyptian economy

Increase in population make heavy load on the Egyptian economy **as Shown in Figure 7,** Egypt Gross domestic product in Egypt was 475.23 billion dollars in 2022 and in 2023 was 393.91 billion dollars and in 2024 it is 347.59 billion dollars, so it is gradatory decreasing that mean population growth has great effectson Egyptian economy that make the government pay more and more to make stabilty in the Economy in egypt and for the Egyptian citizen to obtain his rights however the population growth is increases gradually.

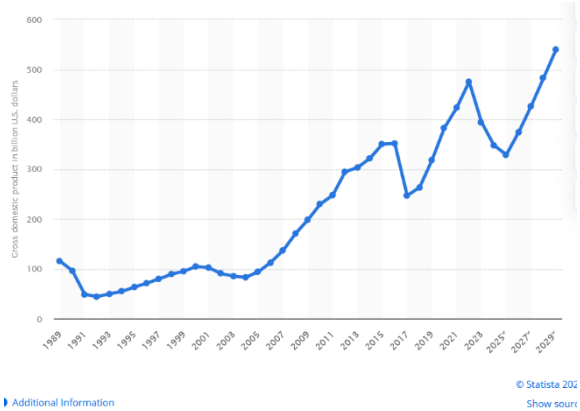


Figure 7 shows Egyptian economy, Statista. (2024c, July 4).

As **shown in Figure 8,** General government expenditure was 40.05 billion pounds in 2022, and it was billion pounds in 2024, which means that Egypt's General government expenditure is increases with a huge rate.

Egyptian borrowing

As the overload on the Egyptian economy Egypt government must make hard choices, such as borrowing in order of meeting the needs of citizens, as the external debt of Egypt is equal 168 billion dollars. After the increase in external dept payments maturing in 2019, in June was 44.4 billion American dollars with 23% from international reserves, and in 2020

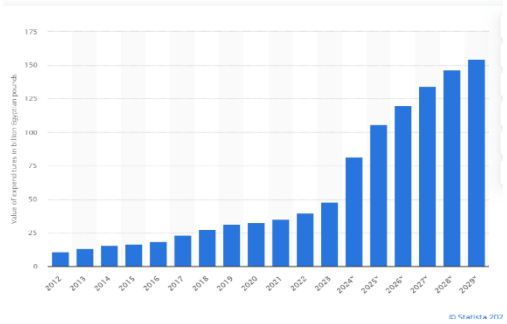


Figure 8 General government expenditure,Statista. (2024, April 23).

in June it was 38.1 billion American dollars, with 26.1% from international reserves, and in 2021, in June, it was 40.5 billion American dollars with 36.6% from international

reserves, and in 2022, in March it was 42.18 billion dollars with 37% from international reserves.

Congestion

as over 100 million people are living in Egypt right now, according to C A P M A S, that makes it hard for the people to live. There will be a lack of products and a housing shortage, as well as racial conflicts, which harm public security in Egypt. Congestion also leads to an increase in unemployment as there are more people and there are fewer jobs**,** which expresses the congestion in Egypt.

Address and reduce pollution fouling our air, water, and soil

table 1 shows cost of health in Cairo 2017

TABLE 2.7: ESTIMATED ANNUAL COST OF HEALTH EFFECTS OF AMBIENT $PM_{2.5}$ IN GREATER CAIRO, 2016/17 (LE BILLION)

	Low	Central	High
Cost of mortality	36.9	38.3	39.5
Cost of morbidity	8.5	8.7	8.8
Total cost of health effects	**45.4**	**47.0**	**48.4**
% equivalent of GDP, 2016/17	1.31%	1.35%	1.39%

Source: Estimates by the author.

Egypt faces many hardships, especially in the types of pollution, which includes water, air, and soil pollution. About 19.200 people died too early, and over 3 billion days were lived with illness in Egypt in 2017 because of nearby PM2.5 air and water pollution in Greater Cairo and all of Egypt. The expected cost of these health effects was equivalent to 1.35% of GDP, about 47 LE billion, **as shown in Table 1.** All these exist due to Air, water, and soil pollution. The major sources of air pollution are transport, agricultural slash and burn, and industrial power. Road transport is the primary source of PM2.5 air pollution in the greater Cairo area (33%). Egypt has reduced this result by halving fuel subsidies, but transport costs have risen enormously, so it has made electric cars.

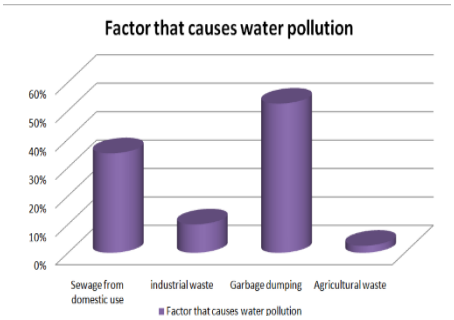


Figure 9 shows factors that causes water pollution

Waterborne diseases have been estimated to cause more than two million deaths and four billion cases of diarrhea annually. Major pollution sources of the Nile and its main canals are effluent from agricultural drains and treated or partially treated as industrial and local wastewater, including oil and waste from passenger and riverboats, which are **shown in Figure 9**. For these causes, EGYPT produced an

awareness campaign and laws by applying several policies to prevent any illegal actions, such as waste disposal and informal construction alone.

The analysis of polluting soil of the Damietta governorate in Egypt showed the highest level of 80% of pollution by different pesticide residues (Hathout et al. 2022) followed by Fayoum Governorate at 40% and Menia at 36.36% then Alexandria at 10.00% Hathout et al. 2022. Causes of soil pollution agricultural activities, Waste disposal, Accidental oil spills, and Industrial activities which **are shown in Figure 10**.

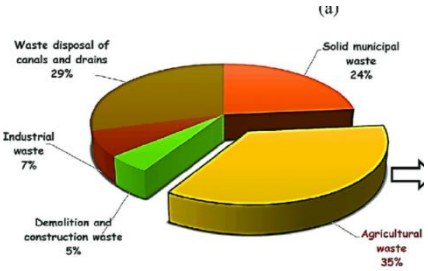


Figure 10 shows the cause of soil pollution in Egypt

Causes

<u>Industrial waste</u>

The Nile River has been subjected to different sources of pollution of through pollution in several complicated ways. Industrial effluent represents a real danger to the Nile River. Egypt is one of the most industrialized countries in Africa which has a high density of factories producing chemicals, fiber, cement, food products, and petroleum-based materials. However, poor waste management systems and the weak effect of environmental laws have led to severe pollution problems. For air pollution in 2017, nearly PM2.5 air pollution in Greater Cairo led to approximately 19,200 early deaths and over 3 billion days lived with illness. The economic cost of these health effects was valued at 1.35% of Egypt's GDP for the 2016/2017 economic year. For soil pollution, the overuse of chemical pesticides in agriculture has resulted in significant soil pollution in Egypt, resulting in approximately 100% loss in some areas. This pollution affects soil fertility and crop yields, and all percentages of them **are shown in Figure 11.**

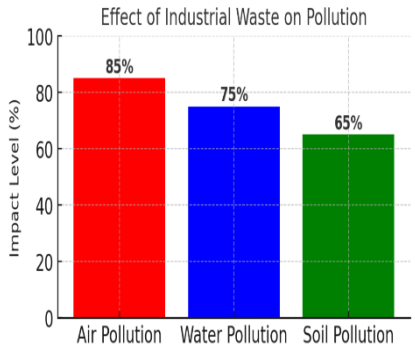


Figure 11 shows the ratio of the industrial waste on the pollutions

Population Growth and Urbanization

The density of industries in cities, especially fossil fuel-dependent cities, has led to increased emissions of sulfur dioxide (SO2), carbon monoxide (CO), and pollutants, **as shown in Figure 12**. In Alexandria, for example, a cement factory converted from natural gas to coal, causing an increased level of domestic air pollution. The development of substructure and residential areas to accommodate the growing population has encroached on green spaces. Cairo lost around 910,894 square meters of green space between 2017 and 2020 because of road widening projects. This loss increases the urban heat island effect and reduces the natural filtration of air pollutants. Burning of Agricultural Waste is producing thick black smoke as the Black Cloud, especially during September to November. This enhances air pollution, increases respiratory diseases, and triggers climate change.

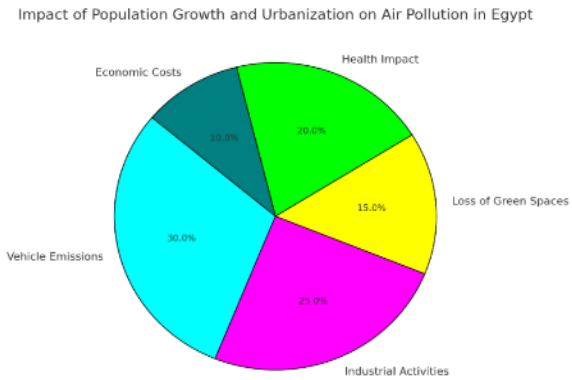


Figure 12 shows the impact of population growth and urbanization on air pollution in Egypt

Impacts

Environmental Damage

Reports say that 38 million Egyptians drink water polluted by sewage, leading to the spread of diseases such as cholera, typhoid, schistosomiasis, and hepatitis. The use of polluted water has been linked to outbreaks of waterborne diseases, such as cholera and hepatitis, all of which impact on the environment. For the Historical Warming Between 1901 and 2013, Egypt experienced an average temperature rise of 0.1°C per decade. This rate accelerated to 0.38°C per decade between 2000 and 2020, higher than the global average increase of 0.31°C, **as shown in Figure 13**. A sea-level rise of 50 cm by 2050 could displace 2 to 4 million Egyptians from the coastal sector, especially affecting cities like Alexandria.

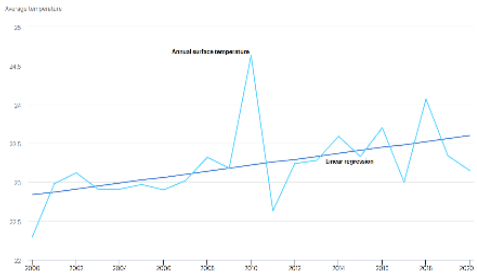


Figure 13 shows the change in the temperature between 2000 and 2020

Economic cost

Environmental pollution in Egypt has a serious economic impact, which affects population health, agriculture, industry, tourism, and total national production. The World Bank calculates that Egypt loses billions of dollars annually due to environmental breakdown. Below is a breakdown of the economic impact of air, water, and soil pollution. Air pollution costs Egypt 3.58% of its GDP annually, which is equal to approximately $47 billion in 2024. For the water pollution, Water pollution costs Egypt nearly $6 billion annually around 0.8% of GDP. Soil deterioration and pollution cost Egypt $3-5 billion annually due to reduced agricultural output. All of them are **shown in Table 2**

Conclusion: Total Economic Cost of Pollution in Egypt

Pollution Type	Estimated Annual Cost	Percentage of GDP
Air Pollution	$47 billion	3.58%
Water Pollution	$6 billion	0.8%
Soil Degradation	$3-5 billion	0.5-0.8%
Total Cost	$56-60 billion	~5-6% of GDP

Table 2 shows the total economic cost of air, water, and soil pollution in Egypt

Marine Life Destruction

Factories near the Nile and coastal areas discharge harmful chemicals, heavy metals, and untreated wastewater into the sea. Egypt produces 5.4 million tons of plastic annually, with much of it ending up in the Mediterranean. Over 74,000 tons of plastic waste pollute Egyptian waters yearly. Over 60% of commercial fish species in the Mediterranean and Red Sea are overfished. For the Red Sea, Egypt's Red Sea coral reefs, which cover 2,000 km², are in danger from rising sea temperatures, which cause coral bleaching and death, and increased CO_2 levels, lowering the pH of seawater, weakening corals and shellfish. Negative impacts from air and water pollution on marine life **are shown in Table 3.**

Category	Impact
Economic Losses	$5-8 billion/year
Plastic Waste	74,000+ tons/year
Overfished Species	60% of fish stocks
Fisheries Decline	$1.5B industry at risk
Tourism Impact	$7B dive tourism threatened
Seafood Prices	10-30% increase
Major Pollutants	Oil, plastic, chemicals

Table 3 shows Economic and Environmental Impact of Marine Life Destruction in Egypt

Recycling garbage and waste for economic and environmental purposes

In a world full of growing waste, the processes of managing and recycling waste are no longer optional but critical, environmentally and economically. Recycling is the process of converting garbage and waste into materials that can be used again to reduce waste and garbage. Furthermore, the annual generation of solid waste is approximately 2 billion tons, which is forecast to expand by 2050 due to population growth. **(Ritchie et al., 2023)**, Also, only 19% of global waste is recycled, with only 37% waste going to the landfill, **as shown in Figure 14.**

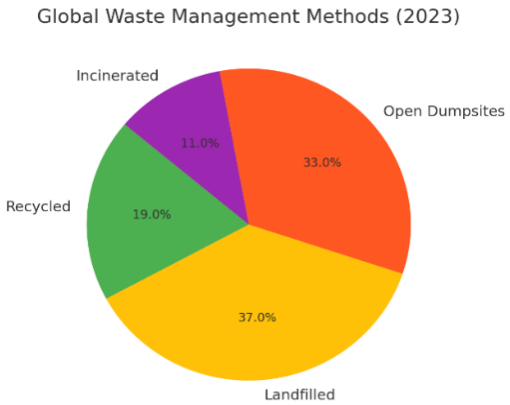


Figure 14: Global Waste Management, Adapted From The UN Environment Program (UNEP) "Global Waste Management Outlook" (2015) Report.

In addition, organic materials usually represent the bulk of the municipal waste generated in Egypt, as shown in **Figure 15**. Thus, organic materials constitute 43.5 percent of the total solid waste generated nationwide. **(Zein, 2021)** Moreover, recycling waste in Egypt has both economic and environmental consequences significant to the government and people, as the annual amount of waste reaches 95 million tons, of which only about 12% to 15% is recycled properly. Also, about 80% of the waste ends up in landfills or informal dumpsites. Improper disposal of the wastes leads to pollution, health risks, and greenhouse gas from decomposition, which is over 5 million tons annually. Egypt can address its waste can of worms by creating a sustainable recycling infrastructure, reducing environmental impact, and building a better economy.

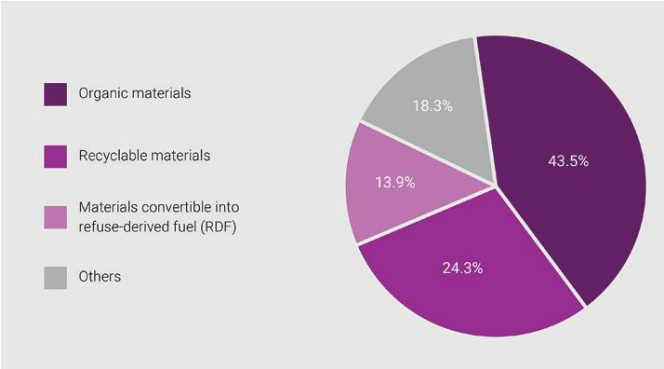


Figure15: Percentage of Recyclable Materials Among Other Types of Waste (EEAA,2021)

Causes

Lack of Governmental Support

Egypt faces a lack of governmental support for recycling garbage and waste. Recycling in Egypt is totally handled by informal sectors, such as general and random dumps. Furthermore, the government doesn't pay cleaners well or hire workers to collect and sort garbage and waste to be recyclable material. Egypt government doesn't provide economic and psychological support for young graduates to work as recyclers.

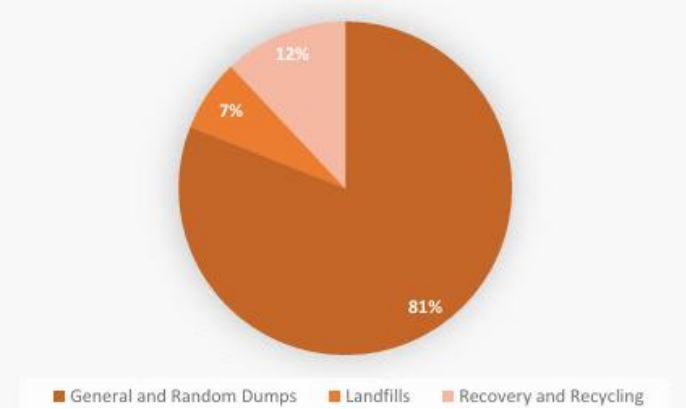


Figure 16: Final Disposal Methods of Waste in Egypt

Additionally, as **shown in Figure 16**, approximately 88% of the waste ends up in landfills or open dumpsites due to a lack of infrastructure. The informal sector is heavily involved in collecting, sorting, transporting and recycling activities. In areas that lack collection services (especially in rural areas) in low-income countries, households manage to get rid of garbage in the nearest empty places, public space or river or even burn it. Also, uncollected waste leads to various environmental and health risks. In addition, it impacts the ecosystems of rivers. (Mahmoud & Ghani, 2017, p. 44) .

Lack of Education and Awareness Campaigns

Modern waste management is virtually impossible without public awareness of the proper handling of waste and the will to tackle and solve this problem **(Miltojević, Krstić, & Petkovic, 2017)**. Moreover, according to the research, both theoretical background and practical findings suggest the crucial need for improving public awareness and encouraging understanding of the consequences of pollution and climate change that are caused by not sustaining the environment. **(Chen, Huang and Lin, 2019)** Research suggests that with dedicated education schemes, recycling rates could be upgraded by up to 30% as people become more aware of the economic and environmental impacts of their waste. Increased volunteering in recycling could help reduce waste management costs by millions annually. Also, a survey shows 35.6% mention that there aren't enough trash cans in the street. **(Attia, 2024)**

Weak Waste Segregation Systems

Egypt's recycling process gets worse because of the dearth of an effective waste separation system at. Most waste is collected as mixed garbage, making it difficult and costly to separate the waste into recyclable materials in a short time. **As shown in Table 4**, studies indicate that nearly 85% of town waste in Egypt is mixed before reaching landfills, reducing the efficiency of recycling performance. Without proper separation at the household and commercial levels, recyclable items like plastics, paper, and metals are often polluted and end up thrown away. Furthermore, making colour-coded bins and collection systems could improve material recycling rates by up to 25%, reduce landfill dependency, and lower greenhouse gas emissions allied with unmanaged waste**. (Abdelaziz, 2023; EEAA, 2024)**

Table 4: Number of Households According to The Disposal Method in 2019

Type	Total number of households	Waste collectors	Dumped in container	Burned	Dumped into canal / Drainage	Dumped into Street	Buried in the ground	Use as Fertilizer	Other
Urban	266741	107609	179180	35941	18810	72267	0	0	190
Rural	735196	230391	210272	354147	201737	197072	37327	89026	3470
Total	1001937	338000	389452	390087	220547	269340	37327	89026	3660

Impacts

Health Risks in the Informal Recycling Sector

Informal recycling can be harmful to the health of workers and their communities in Egypt. Many people working in informal recycling often lack protective equipment and safety measures, exposing themselves to hazardous substances. These materials can include toxic heavy metals like lead and mercury, which can contaminate the air, water, and soil. Materials such as batteries, electronic, and plastic wastes have been commonly found to contain lead up to **200 mg/kg, as shown in Figure 17.** These substances can accumulate in soil and aquatic systems and penetrate the food chain with serious health effects, such as neurological disturbance, kidney failure, and cancer. Children are especially sensitive; they may inhale harmful dust, consume contaminated food, or be exposed to toxins, leading to serious health issues such as impaired cognitive

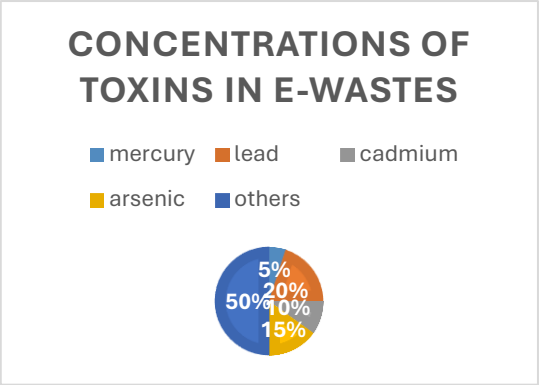


Figure 17: Concentrations of Toxins in E-WASTES

development and increased risks of chronic diseases. Additionally, the informal sector often lacks proper waste management practices, which can exacerbate the pollution and health risks in surrounding communities.

Environmental Decay

Egypt's low recycling rate, around 12 to 15%, contributes significantly to environmental issues. With about 80% of waste ending up in landfills or dumpsites, greenhouse gas emissions from decomposing organic waste are dangerous. Landfills emit large amounts of methane, which is a greenhouse gas 25 times more pollutant than carbon dioxide. improper disposal of toxic waste and plastics has led to pollution in water like the Nile, where an estimated 500,000 tons of plastic waste flow annually. This pollution harms marine and ground ecosystems, threatening living organisms. Additionally, **as shown in Figure 18**, more than 720,000,000 tons of sewage, 142,000 tons of mineral oil, 66,000 tons of mercury, 4,200 tons of lead and 40,000 tons of phosphate all end up in the waterways. Studies (**EEAA, 2021**) show that pollution in Egypt's rivers and soil has reduced agriculture's productivity by 10 to 15% over the last decade. By improving recycling efforts, the country could prevent millions of tons of waste from polluting natural environments and reduce air and water pollution, directly benefiting both ecosystems and public health.

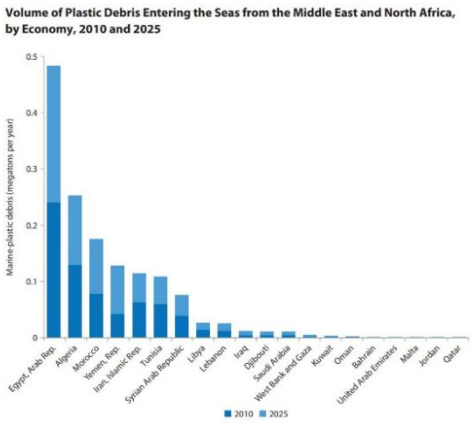


Figure 18: Volume of Plastic Entering The Seas From Middle Est and North Africa 2025

Economic Losses

Egypt's lack of recycling infrastructure results in significant missed economic opportunities. The waste sector could generate revenue from recycling and the production of secondary raw materials. The Egyptian economy loses approximately $5 billion annually due to not enough recycling and waste management practices. For example, if just 30% of Egypt's annual plastic waste, around 1.5 million tons, were recycled, it could generate over $500 million annually in raw material for plastic manufacturing industries. Adjusting the waste management sector could create nearly 300,000 jobs, benefit local economies, and improve ways of living. Without these advances, Egypt remains reliant on imported raw materials for manufacturing, which increases production costs and hinders competitiveness in the global market.

Increasing Industrial and Agricultural Base of Egypt

Countries with strong economies depend on industry and agriculture, which shows the significance of developing these two fields and increasing their bases (Zahoor & Mushtaq, 2023). Some countries have gone through an industrial revolution, transforming their economies from handicrafts into economies based on large-scale mechanized manufacturing, which has led to economic flourishing. The results of this revolution brought about radical changes, bypassing many hurdles and avoiding economic collapse in many countries, for example, Britain and France.

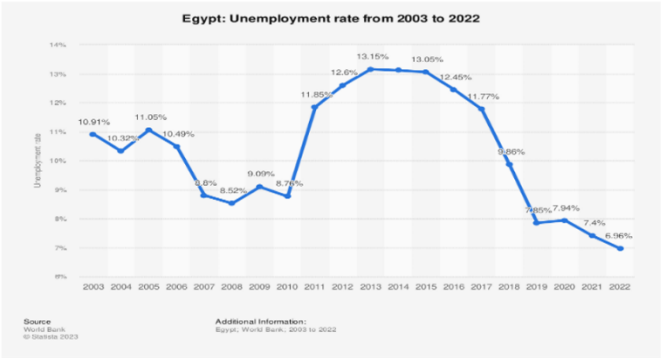


Figure 19: Egypt's unemployment rates from 2003 to 2022.

On the other hand, Egypt faces economic challenges due to the failure of industrial development, which is attributed to political factors. The degradation of the industry has occurred because of the privatization of some factories and companies. Moreover, the importing policy, such as exporting the raw material and importing it after being manufactured, led to a decrease in the process of the industry, and it increased the percentage. Figure 10: Egypt's unemployment rates from 2003 to 2022. of unemployment in the past 20 years, **as shown in Figure 19** (O'Neill, 2023).

Furthermore, Egypt's economy relies on agriculture, as it is considered the fundamental component of the Egyptian economy, contributing to 11.4% of Egypt's GDP and 23.3% of the employment percentage in Egypt, according to the United Nations Development Program (2022). Also, the overuse of pesticides contaminates the soil. All these pollutants affect the ozone layer and may lead to climate change. Climate change is considered one of the biggest problems that will affect the yield and productivity of crops. In addition, the most concerning issue that may destroy agriculture in Egypt is the contamination of water. For example, the main source of water in Egypt is the river Nile, which has become polluted with industrial wastes, pesticides, fertilizers, and plastics. cauis acidic increase in the pH of water, which makes it acidic and toxic. All these issues may cause the deterioration of agriculture, which will degrade the economy and make things worse than before.

Causes:

<u>Lack of exploitation of resources</u>

Firstly, Egypt suffers from a lack of energy because there is no exploitation of the resources, for example, the Mediterranean and Red Seas, which include billions of oil barrels that are not exploited, as Egypt produces approximately five hundred thousand BPD (Barrels Per Day), **as shown in Figure 20**, according to Crude Oil Peak (2022). Additionally, there are hundreds of natural gas fields in the Mediterranean Sea, but the production of natural gas in Egypt depends on a few fields. Also, there are many regions in Egypt, specifically deserts, that haven't been explored well.

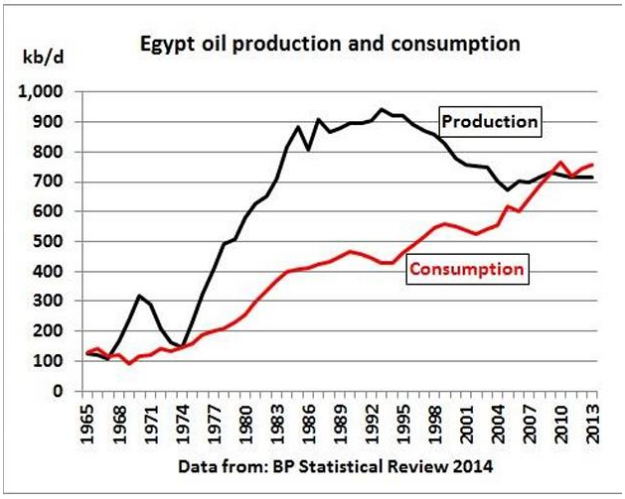


Figure 20: Oil consumption and production in Egypt

However, these areas might contain huge amounts of fossil fuels and other resources. Lastly, this will affect industrial and agricultural progress since 90% of electricity in Egypt is produced by the combustion of fossil fuels, which helps the pumping of water and operation of the factories. Figure 11 Oil consumption and production in Egypt

Water pollution and scarcity**:**

The availability of water is significant for industrial and agricultural growth, as irrigation and agriculture consume 77% of Egypt's water supply, **as shown in Figure 21**. Since it is used in large amounts in the industrial field in many processes, such as cooling systems.

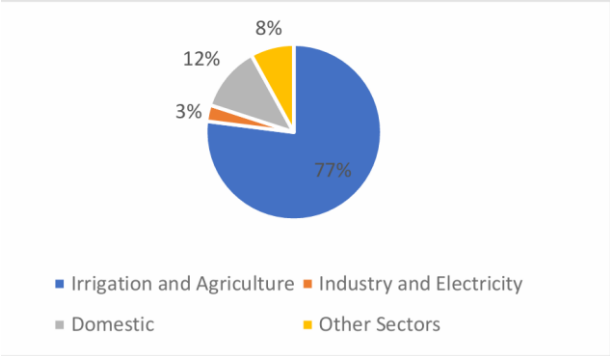


Figure 21: Egypt's water consumption according to sector

Moreover, the contaminated water causes damage to the equipment and poses health risks to the workers, so an inadequate water supply may force the industries to relocate to an area with better resources, resulting in a decrease in employment rates and loss of income. Furthermore, the pollution of water affects the quality and the yield of crops and leads to soil degradation and contamination that decreases land productivity; also, the scarcity of water leads to a decrease in the yield, leading to an increase in food prices (Hejazi et al., 2023).

Lack of governmental regulations and policies

Government regulations play an important role in economic growth. The regulations have a significant impact on agricultural growth, such as the unrestrained use of pesticides and fertilizers, causing soil contamination and harming the natural ecosystem, which leads to lower crop yields. In addition, the lack of government regulations can lead to unfair competition. Large agribusinesses may dominate the market, making it difficult for small-scale farmers to compete. This can lead to the loss of jobs and income for small farmers, as well as a decrease in the diversity of crops growing. Without regulations, in some developing countries, industries may operate without proper safety equipment or training for workers, leading to high rates of workplace accidents and injuries. Also, industries may prioritize profits over safety, as many factories don't follow laws like disposing of industrial waste in clean sources of water. Otherwise

Impacts

Increased dependence on imported goods and resources

As the world continues to become more globalized, countries increase their reliance on the importing policy, in other words, the importing of goods and resources. One of the most significant ways in which Egypt's increased dependence on imported goods and resources is harming agriculture is through the loss of domestic markets, as more products are imported. Local producers are unable to compete, and many are forced out of business. Also, it often results in decreased demand for the locally produced products. This has led to a decline in the availability of locally grown and produced food products, which has in turn had a negative impact on the health and well-being of many people.

In addition, increased dependence on imported goods and resources can lead to a loss of control over the quality and cost of essential goods. For example, if a country relies heavily on imports of essential products such as food or energy, it may be vulnerable to changes in pricing or supply that are beyond its control.

Decline in economic growth and development

The degradation of industry and agriculture can have significant negative effects on the economic growth of a country. When these two sectors decline, it can lead to a decrease in productivity, employment opportunities, and ultimately, economic output.

Industry is a crucial sector for any economy since it contributes to the production of goods and services. However, if the industry degrades due to factors such as lack of investment, outdated technology, or poor regulation, it can lead to a decrease in productivity and efficiency, which can result in higher production costs and lower profits.

On the other hand, agriculture is essential for food security and the provision of raw materials for numerous industries. When agricultural land degrades due to overuse, climate change, or unsustainable farming practices, it can lead to a decrease in crop yield and, ultimately, food shortages. This can lead to higher food prices, reduced export earnings, and a decrease in employment opportunities in the agricultural sector, all of which can negatively affect economic growth, according to the United Kingdom Department for International Development (n.d).

Pollution and health risks

The degradation of industry and agriculture hurts human health. The industrialization process has led to the release of harmful pollutants into the environment, as shown in Figure 13. These emissions cause great health risks, according to the U.S. Agency for International Development (2022). The use of pesticides and other chemicals in agriculture has also resulted **in Figure 22** Greenhouse gas emissions by sector in Egypt environmental contamination, which can lead to health complications.

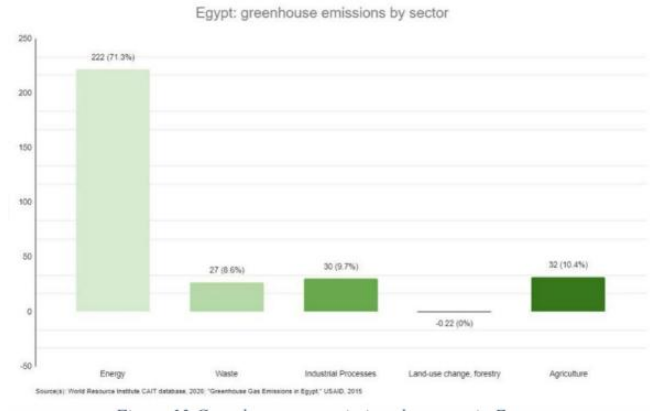


Figure 22 Greenhouse gas emissions by sector in Egypt

In addition, the degradation of agriculture can also lead to health problems. The use of pesticides and herbicides in agriculture has been linked to cancer, developmental problems, and other health risks. Furthermore, the degradation of agriculture can lead to food insecurity, which can result in malnutrition and other health issues. The loss of biodiversity and soil degradation can reduce the nutritional value of crops and limit the availability of fresh produce. In conclusion, the degradation of industry and agriculture can have serious health consequences.

Problem to be solved

Increasing food demand due to population growth places pressure on agricultural systems that rely on limited freshwater resources and unsustainable farming practices.

Irrigation is essential for agriculture, especially in areas with limited rainfall. Various methods like surface, drip, sprinkler, and sub-surface irrigation are used based on crop type, soil, and water availability. Amidst the challenges posed by burgeoning population growth, water scarcity, and environmental changes, efficient irrigation techniques have become increasingly important.

Traditional irrigation techniques frequently lead to substantial water loss through evaporation, runoff, and uneven distribution. This has prompted numerous researchers and analysts to explore alternative approaches, aiming to optimize current methods to boost agricultural output while conserving water resources and ensuring the long-term sustainability of agriculture and water usage.

In an era defined by the urgent need for sustainable solutions, the convergence of agriculture and aquaculture has given rise to a revolutionary method known as aquaponics. This innovative approach not only challenges conventional farming practices but also offers a transformative pathway towards achieving food sovereignty and optimizing water usage.

Within aquaponic systems, a symbiotic cycle unfolds. Fish waste serves as vital nutrients for plants, while these same plants act as natural filters, purifying the water destined to circulate back to the fish tanks. This harmonious relationship between aquatic life and vegetation fosters a closed-loop ecosystem, significantly curbing water wastage and elevating the system's overall sustainability.

Positive consequences

Resource Efficiency & Water Conservation

Water scarcity is a growing global concern, and traditional agriculture is a notorious water consumer. Aquaponics, on the other hand, is a water-efficient marvel, making resource efficiency one of the main advantages of aquaponics.

As mentioned, aquaponic systems utilize a closed-loop design in which water circulates between fish tanks and plant beds. The nutrient-rich water from the fish tanks is used to fertilize the plants, and as the plants absorb these nutrients, the water is naturally filtered, cleaned, and returned to the fish tanks. This efficient recirculation minimizes water usage dramatically, typically using 90% less water than traditional soil-based agriculture.

In regions suffering from drought or where water resources are limited, aquaponics provide a beacon of hope for consistent food production without exacerbating water stress. It's a model of resource conservation that not only preserves water but also reduces the pollution associated with conventional agriculture, where excess nutrients and chemicals often leach into waterways.

Organic & Chemical-Free Produce

Chemical fertilizers and pesticides used in conventional agriculture can introduce harmful substances into our food supply and negatively impact the environment.

Aquaponics offers a natural and chemical-free alternative. Fish waste provides a rich source of nutrients for plants, and the microbial ecosystem within the system helps to break down these nutrients into forms readily absorbed by the plants. This process ensures that the produce is organic and free from harmful chemical residues, making it a healthier choice for consumers.

Reducing Overfishing & Pressure on Wild Ecosystems

Overfishing is a critical global issue, threatening the health of marine ecosystems. Aquaponics offers a responsible alternative by providing a sustainable source of fish. Cultivating fish within controlled environments, it reduces the pressure on wild fish populations. Furthermore, aquaponics enables the farming of species that are not typically found in the wild but are valuable for consumption, thus expanding options for consumers and contributing to biodiversity preservation.

Negative sequences

Large Water Consumption & Scarcity

Nowadays, the agricultural sector consumes 70% of the world's water, **as shown in Figure 23**, withdrawn from aquifers, streams, and lakes, and is ultimately responsible for 13.5% of global greenhouse gas emissions. On the other hand, hydroponics saves up to 95% of irrigation water compared to conventional agriculture. In the case of hydroponics implemented as part of a closed system, the water consumption and nutrient supply are reduced. A study on lettuce yield that compared hydroponics with conventional agriculture showed that the water demands were 20 ± 3.8 L/kg/y and 250 ± 25 L/kg/y, respectively, for this crop.

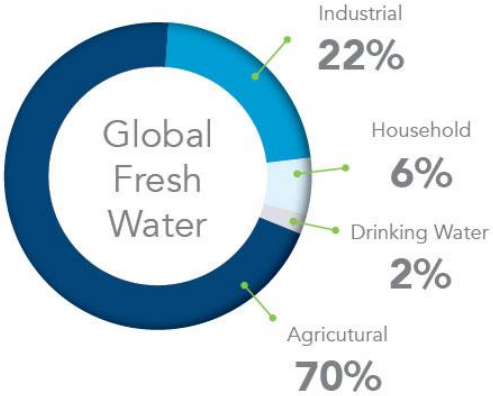


Figure 23 shows *the division of water worldwide*

Soil Degradation and Loss

Conventional agriculture suffers from crucial disadvantages associated with the extensive use of resources, pesticides, fertilizers, and land, all of which must be intensified to meet increasing food production targets. Currently, 38% of non-frozen land globally is at the disposal of crop production, which will rise continuously until 2050, reaching about 593 million hectares of land to meet the food needs of a constantly growing population. The disadvantages of such land use include the destruction of ecosystems and the disturbance of environmental balance

Greenhouse gas emissions

According to *Pomoni et al.* (2023), the agricultural sector is responsible for a significant portion of global greenhouse gas emissions—about 13.5 % to 29 % of total annual emissions when conventional soil-based farming methods are used, largely due to fertilizer use, pesticide application, water pumping, soil disturbance, and energy demands for greenhouse heating and cooling. In contrast, soilless systems like hydroponics (and similarly aquaponics) have lower greenhouse gas emissions: one comparative study cited in the review reported that conventional soil crops produced about 0.23 kg CO_2-equivalent per kg of crop, while hydroponically grown crops emitted about 0.11 kg CO_2-equivalent per kg, which is over 52 % lower emissions.

Research

Topics related to the problem

Rapid Population Growth

Rapid population growth is defined as the significant and often accelerating increase in the number of individuals within a population over a relatively short period of time. Historically, global population growth rates peaked at approximately 2.19% in 1963 and, although they declined to about 1% by 2022, the world population continues to expand by significant numbers each year. This ongoing increase places mounting pressure on natural resources, food production systems, infrastructure, and social services, making rapid population growth a critical challenge for environmental sustainability, economic stability, and long-term development.

High Food Demand

High food demand is defined as the sustained and often rapidly increasing requirement for food resources within a population, driven by factors such as population growth, urbanization, rising incomes, and changing consumption patterns. As demand steps up, it places significant pressure on agricultural systems, water supplies, land use, and food distribution networks. Frequently leading to resource exhaustion, higher food prices, environmental degradation, and increased risk of food insecurity. When food demand grows faster than production capacity or supply-chain efficiency, it can contribute to shortages, unequal access to nutrition, and long-term economic and ecological instability.

unsustainable farming practices

are defined as agricultural methods that damage the environmental resources or reduce long-term productivity by prioritizing short-term yields over ecological balance and resource conservation. These practices may include excessive use of chemical fertilizers and pesticides, over-irrigation, monocropping, deforestation, and poor soil management techniques. Over time, such approaches can lead to soil erosion, nutrient depletion, water shortage, biodiversity loss, pollution, and reduced resilience to climate change. As these impacts accumulate, unsustainable farming threatens food security, farmer livelihoods, and ecosystem stability.

Topics related to the solution

Aquaponics system**.**

Aquaponics is a system that combines the plant and the fish in a closed water loop. There are many ways to apply it, like NFT, RAFT, DWC, aeroponics, and drip systems.

This concept helped us to choose and apply the correct solution, and after researching, we know any struggles that we may face.

Biofilter for an aquaponics system.

A biofilter means the tank that contains beneficial bacteria, which are stuck on the surface of small plastic pieces.

This stage is imperative for the water cycle in this system because the bacteria convert fish waste (ammonia) to useful nutrients for the plant. So, learning about this concept helps us to design the biofilter tank

Manner of Choosing Plant and Fish.

Aquaponics is a delicate balance of fish and plants coexisting in a symbiotic system. The success of an aquaponic garden depends on selecting compatible fish and plants that thrive under similar environmental conditions. Understanding the relationship between these two crucial components ensures a healthy and productive system.

Matching fish and plants correctly is essential for a successful aquaponic garden. And find the setpoints of the plant and fish. By considering water temperature, pH levels, nutrient availability, and waste production, you can create a balanced ecosystem that supports both aquatic and plant life.

Measuring parameters.

Any successful system needs to be constantly observed for environmental parameters. There are many essential variables to be monitored in an aquaponic system that help to keep the plant and fish alive, and provide the best circumstances for plant and fish growth

These parameters are temperature, water level, and light intensity.

Other Solutions Already Tried

Upward Farms

Figure 24: Upward Farms

Upward Farms is a new technology regenerating an aquaponics farm. It's a 250,000-square-foot facility in Luzerne County, Pennsylvania, which is the world's largest indoor vertical farm, as shown **in Figure 24**. Also, spanning approximately six acres, this huge farm is 2 or 4 times the size of its nearest competitors. Founded in 2013 as Edgeworks by Jason Green, Ben Silverman, and Matt LaRosa. The project represents a major shift toward localizing the production of highly short-lived foods, specifically USDA Certified Organic microgreens and Best Aquaculture Practices-certified hybrid striped bass. Environmentally, the farm is a model of conservation, designed to save more than 100 million gallons of water and 120 acres of land per year. **(Al-Kodmany, 2018)**

<u>Mechanism</u>

Figure 25: Upward Farms Project structure

The farm operates as a complex, self-regulating biological manufacturing platform. The mechanism is built on three pillars: Biological Symbiosis, Ecological Intelligence, and Advanced Automation, as shown **in Figure 25**. The core mechanism is a closed loop aquaponic system.

Firstly, the process begins in large indoor tanks where hybrid striped **bass** produce waste in the water, which contains nitrogen and other nutrients essential for plant growth. Beneficial bacteria naturally break down fish waste into forms that plants can absorb. Upward Farms intentionally curates a diverse community of **bacteria and** fungi (the microbiome) within the system, which acts like a natural immune system. They use sensors and computer software to constantly monitor fish and plants. If the system detects an abnormality, it automatically adjusts the water flow, temperature, or feeding

schedules on its own. Lastly, the farm is fully indoors and climate-resilient, which enhances the farm's effectiveness. Furthermore, it uses custom LED lighting designed to emulate the natural solar spectrum so as not rely on the sun. The company uses a proprietary app to track operations data, including pH levels, water temperature, humidity, and building temperature. The agronomy team analyzes this data to identify the exact conditions that lead to successful growth cycles and replicates them to ensure consistent quality.

Points of weakness:

1. Energy and Infrastructure Costs:
   Traditional farming gets sunlight, rain, and airflow for free. Upward Farms had to pay to artificially recreate all those things 24 hours a day, 7 days a week.

2. Biological Complexity:
   Hydroponic farms only have to worry about keeping plants and fish alive. Upward Farms had to manage three distinct living systems at the same time: the plants, the fish, and the microbiome

3- Limited Crop Viability:

Because it costs so much money to run this high-tech farm, Upward Farms had to sell crops that grew incredibly fast and sold at a very high price.

Points of Strength:

1. Massive Productivity:
   The facility achieves 2 times the industry average yields while eliminating 99% of crop disease through its focus on biodiversity and microbiome.

2. Freshness Advantage:
   By localising production, the farm delivers produce within 24 hours of harvest, making it up to 40% more nutrient-dense than greens come from the West Coast.

3. Sustainability and Certifications:
   It is the only large-scale vertical farm providing USDA Certified Organic greens and Best Aquaculture Practices-certified fish, saving 100 million gallons of water annually.

Cylburn Arboretum (Baltimore, MD, USA)

Aquaponics is a form of aquaculture that integrates soilless crop production (hydroponics) to raise edible plants and fish, **as shown in Figure 26**. The fish are fed and excrete waste, which is broken down by bacteria into nutrients. Plants utilize some of these nutrients, and in the process, filter the water in the system. Most aquaponics systems are recirculating aquaculture systems where water is continuously recycled through an interconnected series of fish tanks and waste treatment systems (Timmons and Ebeling, 2002). Early attempts at recirculating aquaculture were challenged by the accumulation of ammonia, a potentially toxic by-product of fish waste (Bohl, 1977; Collins et al., 1975).

Figure 26 shows Cylburn Arboretum aquaponics

In one approach to improve water quality, researchers incorporated plants as biofilters (Lewis et al., 1978; Naegel, 1977; Sneed et al., 1975; Sutton and Lewis, 1982), which was an early application of aquaponics. The objective of this study was to describe the operating conditions, inputs (energy, water, and fish feed), and outputs (edible crops and fish), and their relationship over two years for a small-scale raft aquaponics operation in Baltimore, Maryland, United States. The system had roughly 1% water loss per day and used an average of 35,950 L for replenishment per year. Predicted values suggest rainfall could completely replace the existing water needs.

The average energy use was 19,526 kWh for propane and electricity per year at a cost of $2055 US dollars. The largest uses of electricity were in-tank water heaters. Comparing inputs to outputs, 104 L of water, 0.5 kg feed, and 56 kWh energy ($6 in energy costs) were needed to produce 1 kg of crops; and 292 L of water, 1.3 kg feed, and 159 kWh of energy ($12 in energy costs) were needed to produce a 1 kg increase in tilapia. Raising tilapia was a net loss, while raising crops was a net gain when comparing market prices to energy costs. Understanding energy, water, and feed use in aquaponic systems is essential to inform farm business plans.

Mechanism

The 10.3 m^3 aquaponics system was sited in a 116 m^2 hoophouse on the grounds of the Cylburn Arboretum in Baltimore, Maryland, U.S. The system was operated with fish and plants for six months (starting in June 2012) before the study period began to allow the biofilter to ripen and nutrient levels to increase sufficiently to support consistent crop growth. The period under study was January 1, 2013, to December 31, 2014. The system's design and specifications are presented **in 27**. Four fish tanks were part of the same system and should be considered one experimental unit. It is typical for aquaculture systems to have more than one tank so that fish at varying stages of development can be raised and harvested in a staggered fashion.

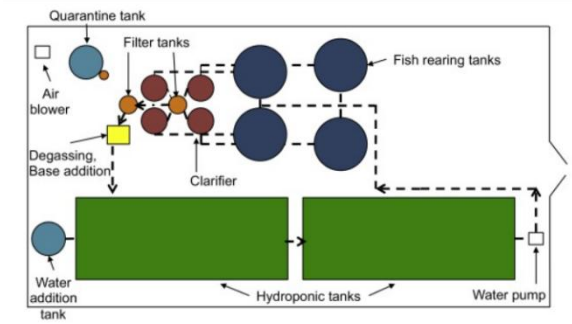


Figure 27 shows Small-scale raft aquaponics system layout

Mechanical components drawing electricity were a water pump, an air blower, four in-tank electrical water heaters, a 4-ft wall-mounted greenhouse fan, an inflation blower to maintain a pillow of insulation between the layers of greenhouse film, several box fans to distribute air throughout the greenhouse, and fluorescent lights. In cold weather, thermostat-controlled, propane-fired space heaters maintained the air temperature at no less than 4–7 °C. If the water temperature dropped below 22 °C, the thermostat-controlled electric heaters operated.

The system did not have an electric water-cooling mechanism, and in the summer months, the water temperature would increase above 22 °C. To mitigate excessive temperature increases, in the summer months, a 50% shade cloth (Aluminet, Maryland Plants and Supplies, Inc.) was installed above the hoophouse, a reflective plastic tarp was hung 1.5 m above the fish tanks, and a thermostat-controlled 4-ft greenhouse fan was used to pull air through two sets of louvered windows. Additional cooling was achieved by rolling up the sides of the hoophouse to 1 m in height. In the event of a power outage, backup power was supplied by a propane-driven generator.

Fish tanks were stocked with 21 Nile tilapia (*Oreochromis niloticus*) to ripen the system, and 227 blue tilapia (*Oreochromis aureus*) were stocked for grow-out. For the

first year of the study period, the fish were fed two different plant-based diets: for 9 months, a slow-sinking feed with 50% protein provided by Watson et al. (2013), and for 3-months an expensive and less palatable USDA Organic feed with 32% protein (AquaOrganic diet from The Aquaponics Source).

Point of Strengths

Water use

Water was continuously cycled through the system at a rate of 93 Lpm throughout the study period. Water additions were made from a 625 L storage tank into the aquaponic system. The storage tank allows for a waiting period in which chlorine can dissipate from the municipal water supply, which can then be gravity-fed into the hydroponic tanks. Sources of water loss were evaporation, evapotranspiration, spillage, leakage, and water exchange (38 L of 10% fish solids per day). After experiencing significant condensation during the winter months on the interior of the greenhouse film, additional measures were taken to cover the fish tanks in the winter using a radiant barrier (TekFoil) to reduce the heat and water loss due to evaporation out of the top of the tanks and reduce the relative humidity in the hoophouse.

Water quality and chemical amendments

Water treatment was performed using four 190-L cone-bottom clarifiers (one per fish tank) followed by two 132-L biofilter tanks in series (the first one filled with orchard netting, the second filled with aerated Kaldnes K1 media). Data on water quality monitoring and the addition of chemical amendments are presented **in Figure 28**. Alkalinity and pH were maintained using a combination of hydrated lime (Good Earth Organics, Cave Junction, OR) and potassium bicarbonate (Nuts.com, Cranford, NJ), typically daily in 50–150 g amounts.

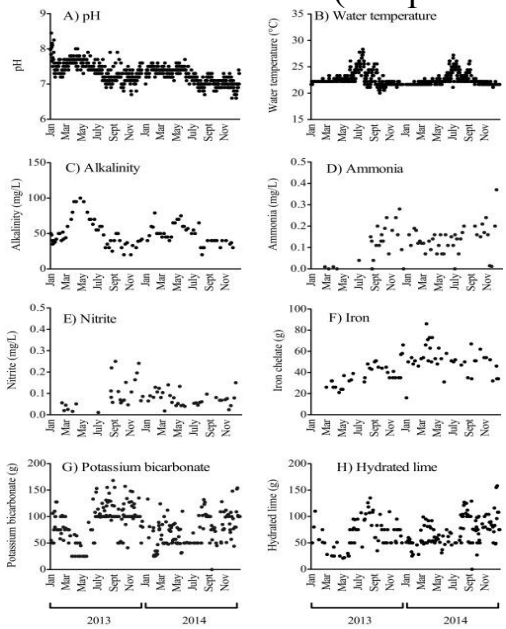


Figure 28 shows Water quality monitoring (A) pH, (B) water temperature, (C) alkalinity, (D) ammonia, (E) nitrite, and chemical amendments (F) chelated iron, (G) potassium bicarbonate, and (H) hydrated lime over two years in a small-scale raft aquaponics system.

The water pH was measured daily using a colorimeter (model DR850, Hach, Loveland, CO) and a phenol red indicator (Hach). The alkalinity of the system water was measured weekly (Lamotte, Chestertown, MD). Ammonia and nitrite were

measured routinely using a colorimeter (Hach) to confirm that levels were below known tolerances for tilapia. Powdered iron chelate (Grow More, Gardena, CA) was added in roughly 20-80 g amounts on a semi-weekly basis to the base addition tank whenever yellowing was observed in plant leaves. Greenhouse temperature and water temperature were measured daily using a greenhouse thermometer and an in-tank digital thermometer. Dissolved oxygen was not measured routinely.

Fish and plant production

A sample of five to ten fish from each tank was weighed on a semi-monthly basis (13 times over the course of the study), and these weights were multiplied by the estimated number of fish in a tank to calculate a tank weight. Samples were obtained by dipping the net into each tank five times and weighing the fish caught.

A variety of crops were grown in the aquaponics system and harvested on a weekly basis starting in January 2013, and sold at market rate at a local farmers' market. Weights of the harvested portion of each crop were recorded. Seasonal plantings included herbs (basil, chives, dill, etc.), loose-leaf greens (salad mix, mustard mix, etc.), greens harvested by the head (lettuces, bok choi, tat soi, etc.), fruiting crops (cucumbers, peppers), and bunched greens (kale, chard, mustard).

Point of weakness

Energy use

There were five main contributors to electrical energy usage in the system: water heaters, air blowers, box fans, a pump, and lights in the winter for seedling germination. Propane was used to power a backup generator in case of a power outage and heaters to ensure that night-time temperatures in the hoophouse remained above 7 °C. Data for electrical and propane usage were collected from the site. Energy calculations for the radiant solar heat gain (RSHG) and solar heat gain factor (SHGF) (American Society of Heating, 2013) were estimated based on the orientation of the site (Google Maps, Mountain View, CA), the specific latitude, the cloudiness for the zip

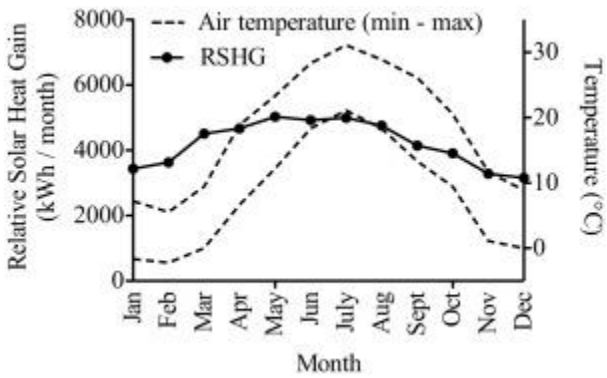


Figure 29 shows the relative solar heat gain (RSHG) in the hoophouse accounting for thermal losses, and average monthly max and min air temperature in Baltimore, Maryland in 2013.

code, and estimates of infiltration and exfiltration of outside air due to the prior construction effectiveness of the hoophouse

Hoophouses are designed for passive solar heating, and the total annual RSHG for the orientation of the hoophouse and its latitude was predicted to be 71,924 kWh, which does not account for shading from nearby trees. When accounting for losses due to the hoophouse building envelope, the total effective RSHG annually was determined to be 50,443 kWh, and reported by month **in Figure 29.**

Biomass relative to water and energy

Table 5 presents the monthly inputs of water, feed, and energy required to produce 1 kg of crops in a small-scale raft aquaponics system. An average of 104 L of water, 0.5 kg of feed, and 56 kWh of energy was needed to produce 1 kg of crops. The most efficient season for converting feed to plant biomass was the spring, where 1 kg of fish feed was converted into 5 kg of plants, which were mainly leafy green crops such as chard, lettuce, and kale. Others have reported as much as 9 kg of lettuce grown using 1 kg of fish feed (Love et al., 2015).

Table 5 presents the monthly inputs of water, feed, and energy required to produce 1 kg

Metric	Average	Month									
		Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct
Water (L) per 1 kg crops											
2013	125	211	189	54	91	113	167	273	175	129	214
2014	83	70	88	43	65	43	113	105	155	184	98
Average	104	140	139	48	78	78	140	189	165	156	156
Fish feed (kg) per 1 kg crops											
2013	0.4	0.6	0.5	0.1	0.2	0.2	0.4	1.2	0.7	0.7	0.9
2014	0.5	0.8	0.6	0.4	0.2	0.2	0.4	0.4	0.7	1.0	0.6
Average	0.5	0.7	0.6	0.3	0.2	0.2	0.4	0.8	0.7	0.8	0.8
Energy (kWh) per 1 kg crops											
2013	66	668	165	61	37	15	17	19	7	10	59
2014	47	340	96	71	30	10	3	3	16	21	37

The average monthly energy cost to grow 1 kg of crops was $6, and varied by month, with a maximum of $55 in January to a minimum of $1 in May through August. In a study of the economics of small-scale aquaponics in Hawaii, average energy costs were $0.73/kg of lettuce, which were similar to the costs we observed during summer months (Tokunaga et al., 2015). At the farmers market where our crops were sold, loose-leaf salad greens sold for $26.52/kg and bunched greens (such as kale and chard) sold for $8.82/kg, which shows that for most months, sales were greater than energy costs when selling at farmers markets.

Superior Fresh

Central Wisconsin is famous for its dairy farms, but it's also home to the world's largest aquaponic greenhouse, spanning over 13 acres. Superior Fresh is a family-run farm growing sustainable salmon and organic leafy greens for consumption across the Midwest and beyond.

Salmon is the most commonly eaten fish in the United States. In 2020, the United States imported over 637.1 million pounds of salmon from abroad, the vast majority of which is farmed and transported by air freight. Superior Fresh is doing their part to reduce this seafood deficit. All their salmon is transported over land, with a smaller environmental footprint, providing Americans with a healthy source of domestic salmon.

Steve Summerfelt, lead scientist at Superior Fresh, is the mastermind behind raising salmon from egg to market. "Our salmon is fed organic-certified feed and is certified non-GMO by a Greener World, with a 'heartcheck healthy' certification from the American Heart Association," said Summerfelt. "Atlantic salmon is a superfood. High levels of OMEGA-3 fatty acids do incredible things for heart and brain health."

At Superior Fresh, salmon are grown on land using freshwater tanks. It is a nearly closed system, with 99.9 percent of water treated and recycled within the facility. Fish waste is put to good use through a nutrient recycling process. Using a microbial biome that mineralizes the waste and releases its nutrients, the waste is repurposed and is used to fertilize the leafy greens. In addition to rigorous hygiene practices, a series of filters and disinfectants are employed to keep food clean.

Mechanism

It depends on the decoupled aquaponics, and this system consists of two rings or parts. Firstly, the fishing system is called RAS Recirculating Aquaculture System, and this system provide or focus on the high intensity and the quality of fish. And secondly, there is Hydroponic, also called water Deep culture, and this works by making the plant absorbed their nutrients, which is in the origin is bad things or matter for the water, and these nutrients and the structure make the plant grow rapidly.

•Block diagram description of the water flow from fish tanks and passes through mechanical filters, bio interactions, and after that, the chemicals go to fish tanks.

•System workflow entrance smolts in the tank and seeding. Fish feed on particular things. After that, fish excrete metabolic wastes such as NH3, which are pollutants for the water, so we get rid of them by seeding, which is added. But the plant doesn't get these by this form, there are chemical bacteria in filters which can transform NH3 to NO3.

•Drum Filters do mechanical streaming, and this removes solid particles, this prevent decaying, and we will store the percentage of the oxygen.

- The software of the farm depends on three parts: inputs from sensors, preprocessing, and outputs. Sensors follow and store the data in SQL. After that, this data flows through PID-Control. This system depends on comparing the current reading with the steady and the best state for the State which be on it. And according to this process, the actions will happen like open oxygen pump or sensor heating.

- Growth models that can predict the best date for the complete growth of a type of fish in the fish tanks, and this according to different variables.

And they are part of the lighting system by providing the best light spectrum for the environment of the fish.

Point of strength :

Climate independence:

the ability of the project to mentain the suitable temperature in this region however the tempereture in this area approach -30 and this refers to the engineering and the effecient of heating system. The defence system and sustainablity which face the seasonal bacteria, fungi and etc.

Symbiotic efficiency:

use the plant secretions to be neutrints for the plant and this mean 100% self-independence and this leads to much decreasing in using of the chemical components. And this save the environment from the dangerous and bad side effect of using these fertilizers.

Nutritional Bio-fortification:

The control on the quality of the water and system providing which work by automation and has huge software system provide and improve the farm lead to increase the quality of products and this result to increase in economic and make good goods for expoting. In addition the rate of growth increase by 30% and this leads to high quality, big production without any bad side effect.

Zero Discharge(sustainablity):

Recycling of the water which approach 99.9% in this huge area (800 acre) is incredible. And this encourage investing in this project and get funds and many partnerships from different and big companies

Points of weakness

High Cost:

Building this system or project costed much money and this result in needing of high funds and investments this project costs about 50 million dollars.

And the back of profit will take a long time from this project, and the system of maintenance costs a lot too. So the improvements in this project and trials of widness it is impossible and need much time, money, and external factors; maybe you will lose control of it.

•Energy Vulnerability

This project depends on the highest amount of energy and any error or mistake in this part will lead to disaster can damage everything. Pump oxygen, filters, software system, etc., all of that depends on the energy.

•Biological Fragility:

Because the project depends on the mutual work of the (plants and fish) so any mistake in each one can stop working whole system so you will need a big staff to follow and fix the system. And no any one can work in this farm because it needs advanced engineers and workers to treat with this complex tools and system.

Aquaponics Farm Al-Bustan

At the northern edge of Cairo, close to the main road to Alexandria, Egypt's first commercial aquaponics farm is located, Bustan Aquaponics, as **shown in Figure 30.** The owner, Mr. Faris Farrag, applied the concept that was developed by Dr. James Rakocy of the University of the Virgin Islands. In the Bustan Aquaponics farm, vegetables and fish are part of one integrated recirculation system. The vegetables benefit from the nutrients provided by the fish through a nitrification cycle that converts toxic ammonia from fish waste into nitrate, making the water suitable again for the fish and highly nutritious for the plants. The construction of the Bustan Aquaponics farm started in 2011.

Figure 30 Bustan Aquaponics

The farm is self-funded by the owner, and investments to build it were in the range of 300,000 Egyptian pounds (approx. US $ 50,000). Work on the farm is done by the owner, the farm manager, and one laborer. A caravan near the farm site accommodates the office and a separate sleeping quarters for the on-site manager. The aquaponic farm sits on 1000 m2 of land within an eight-acre olive orchard and consists of two greenhouses made of fine-meshed netting material. The larger greenhouse is the horticulture unit, containing a deep channel raft system, consisting of six 30 m long basins with XPS polystyrene foam plates floating on the water surface, on which the vegetables and herbs are growing. In the smaller greenhouse, the fish tanks and water treatment system are placed. The greenhouses are located 3 m apart and are connected through 2 underground tubes through which the water flows from the fish unit to the horticulture unit and back again.

Its system combines fish farming and soilless plant cultivation in a closed-loop water system. creating a continuous recycling cycle, which produces pesticide-free Nile tilapia fish, four varieties of lettuce, baby spinach, purple kale, Swiss chard, celery, pak choi, wild rocket, garlic, thyme, and herbs like basil, mint, and parsley. These products are distributed to supermarkets, restaurants, and customers in Greater Cairo.

Mechanism:

The raft system is an aquaponics system that uses a deep-water culture method to grow plants. In this system, plants are grown in floating rafts on top of a tank filled with nutrient-rich water where fish are raised, **as shown in Figure 31**. The roots of the plants are suspended in the water, and they absorb the nutrients they need to grow directly from the water. The raft system is sometimes also called the deep-water culture (DWC) system due to the deep water in the tank.

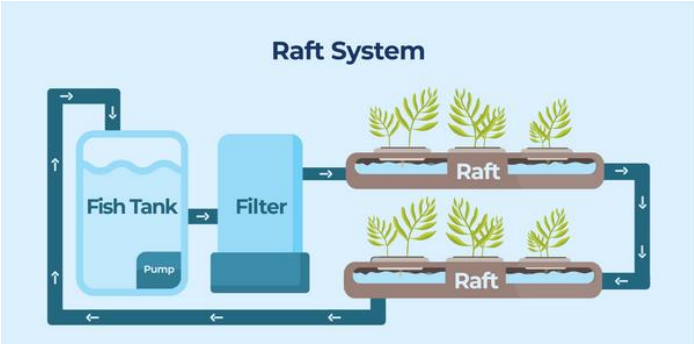


Figure 31 The mechanism of Raft system

The raft system is designed to create a symbiotic relationship between plants and fish. The fish provide the plants with the necessary nutrients for growth; in return, the plants clean the water for the fish. The water is filtered through the plant's roots, which remove excess nutrients and other harmful compounds from the water, creating a healthy environment for the fish to thrive.

One of the defining features of the raft system is the use of a floating raft. The raft is typically made of foam or a similar material that is buoyant and can support the weight of the plants. The raft provides a stable platform for the plants to grow on while allowing easy movement of the rafts for maintenance and harvesting.

The raft system is a highly efficient and sustainable method of food production that utilizes a symbiotic relationship between fish and plants. It offers several benefits, including high productivity, low water use, and a year-round supply of fresh vegetables and fish.

Points of strength:

Saves a large amount of water compared to traditional farming

This farm can save a substantial amount of water when using aquaponic farming since aquaponic systems can effectively recirculate water for lower consumption. Farmers typically use too much water in traditional farming because a large portion of it is lost before reaching its intended destination. Along with water evaporation, water can be lost during field farming when it puddles or rolls away. It's important to understand that the usage of water is 90% less than traditional farming practices. This farm can help to lower this consumption; thus, it can benefit the environment.

Designed for desert conditions.

This farm is also designed to be ideal and suitable for the desert environment and conditions, which means that there are many measurable parameters to ensure the water cycle. This is designed by applying many strategies, using water efficiently, choosing plants and animals that fit the climate. These factors lead to making the farm thrive in a harsh.

Requires no soil and reduces chemical fertilizer use.

This farm also helps to use fewer chemicals when compared to traditional forms of agriculture. Even though pest issues can still be a problem with aquaponic farming, these issues occur on a more infrequent basis, which means that you won't be required to use as many herbicides and pesticides. Since aquaponic systems are controlled tightly, you shouldn't have a problem with weeds. The lack of weeds to kill means that you should be able to avoid the use of herbicides. Keep in mind that the majority of aquaponic systems are kept indoors, which makes it more difficult for pests to get into these systems.

Points of strength:

High initial setup and equipment cost.

Aquaponics systems usually cost more to set up than traditional farming because you need to install all the equipment before starting production. The system requires fish tanks, grow beds, water pumps, filters, and monitoring devices to keep water conditions suitable for both fish and plants. Buying and installing this equipment can be expensive, especially for beginners or small farmers. Since the system depends on this equipment to function properly, the cost cannot be avoided, and farmers need to invest money first and wait some time before they start getting returns from selling fish or crops.

Needs technical knowledge.

Aquaponics requires some technical knowledge because the farmer has to take care of both fish and plants while keeping the water in good condition for both. The person running the system must understand how water quality affects fish and how plants get nutrients from the water. They also need to know how to operate equipment like pumps and filters. If one part of the system is not working correctly, it can harm both the plants and the fish. For this reason, aquaponics can be difficult for beginners who do not yet have experience with this type of system.

Chapter 2: Generating and Defending a Solution

Solution Requirements

System Configuration Requirements

- Use a Deep Water Culture (DWC) raft system where plants grow on floating rafts.
- Polystyrene (EPS) or food-grade foam rafts with plant holes (net pots).
- Continuous water flow under rafts to prevent stagnation.
- Separate fish tank, mechanical filter, biofilter, and raft grow bed.

Plant Selection Requirements

- Best crops for raft systems:
    - Lettuce
    - Basil

Sustainability & Weakness Mitigation

To solve known raft-system weaknesses:

- High oxygen demand → redundant aeration
- Power dependency → backup power
- Disease spread in shared water → strict biosecurity
- Nutrient imbalance → controlled feeding rate & plant density

Energy & Pumping Requirements

- Continuous water circulation pump (24/7).
- Air pumps for oxygenation (critical).
- Energy efficiency preferred (low-head pumps).
- Emergency power option (generator).

Design requirements

The Scale of The Prototype

- The prototype will be a small-scale, functional unit—either a maximum 10 L aquaculture system or a 0.25 m^2 crop growing area
- **Test plan**
- The prototype will be capable of operating continuously for at least 6 hours without manual intervention. The prototype must be portable, testable, and fully operational on the day of the exhibition.

Construction Materials:

- The prototype must be constructed using waste or recycled materials.

Controlled Variables and Setpoints

- The team defined three
- environmental parameters (temperature, water level, and light intensity)

Actuator selection

- Heater, DC fan, LED, water pump, and sump.

Natural Materials Preference (where appropriate):

- The body of the prototype (except for electronic components) will be constructed using recycled
- or waste material, like plastic, and Pipes.

Plant selection**:**

The team chose plants that grow in different

- environmental conditions than those present in your local school area, such as lettuce and basil.

Selection of solutions

Our solution is an aquaponics system. Aquaponics is a revolutionary approach to gardening that combines aquaculture (raising fish) with hydroponics (growing plants in water). The result is a self-sustaining ecosystem where your fish feed on the plants, and the plants in turn, keep the fish tank clean. It's efficient, eco-friendly, and perfect for anyone interested in sustainable gardening.

How Does Aquaponics Work?

While it may sound complex, aquaponics is just a carefully balanced ecosystem. Here's a beginner-friendly breakdown of how it all works:

Step 1: Fish Provide Nutrients

Fish produce waste in the water, which contains nitrogen and other nutrients essential for plant growth.

Step 2: Bacteria Convert Waste into Plant Food

Beneficial bacteria naturally break down fish waste into forms that plants can absorb. Think of them as your microscopic garden helpers!

Step 3: Plants Filter the Water

As plants take in nutrients, they clean the water, which is then returned to the fish tank, creating a self-sustaining loop

Raft System of Deep-Water Culture (DWC)

Also known as raft aquaponics, DWC systems float plants on a sheet of foam over deep troughs filled with nutrient-rich water, **as shown in Figure 32**. The roots dangle directly into the water, absorbing nutrients continuously.

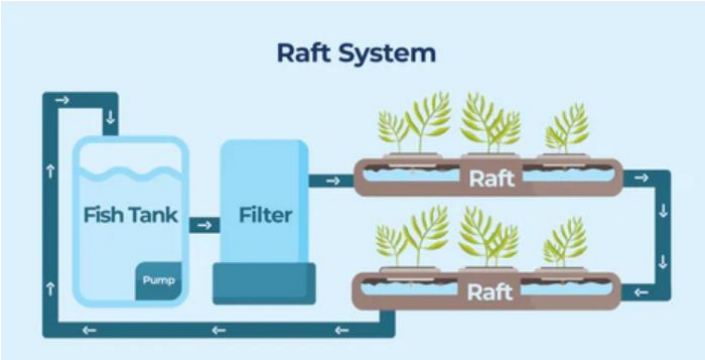


Figure 32 shows the shape of the DWC

How it works:

- Fish waste is filtered out before reaching the grow bed.
- Clean, oxygenated water flows through the raft bed.
- Air stones or diffusers are used to oxygenate the water to prevent root rot.

Benefits:

- Highly scalable for commercial systems
- Ideal for fast-growing crops like lettuce, kale, basil, and chard
- Lower maintenance once established

The type of fish

Tilapia

The type of plants

lettuce and basil

Benefits of Hydroponic Lettuce

- Year-round growth, faster harvests (~30% quicker than soil).
- Water-efficient (up to 90% less water).
- No soil-borne pests, often pesticide-free.

Herbs (Flavor Powerhouses)

Herbs grown in DWC are often more aromatic and potent than their soil-grown counterparts.

- **Basil:** The king of DWC herbs. It grows into a massive, bushy plant. Genovese, Thai, and Lemon basil all perform exceptionally well.

Hydroponic Basil Benefits & Key Features

- **Faster Growth & Higher Yields:** Plants get optimal nutrients and oxygen directly to their roots, leading to faster growth cycles and potentially higher harvests per square foot compared to soil.
- **Water Efficiency:** Hydroponic systems recirculate water, using up to 90% less water than traditional soil farming.

Our location

We chose Borg El Arab City in Alexandria government to be our real-life location, **as shown in Figures 33 and 34**, due to its environmental and demographic characteristics that contrast with our school's location. Unlike the arid desert climate of October Gardens, Borg El Arab features a coastal environment with an average humidity of 70% and a moderate, stable climate year-round. This selection allows us to address the challenge of increasing the industrial` and agricultural base of Egypt. Furthermore, as a part of the Alexandria government, which houses 5.5 million of resident which addresses the Egyptian Grand Challenge of dealing with population growth and its consequences. The city provides a broad land availability for scalable agriculture, serving as a strategic hub to increase Egypt's agricultural base.

Figure 33 shows Borg El Arab

Figure 34 shows the beach of borg El arab

Dimensions

The prototype will be designed as a small-scale, functional aquaponics unit, meeting the specified constraints of either a maximum 10 L aquaculture system or a 0.25 m² crop growing area.

Fish Tank Dimensions

- **Shape:** Rectangular
- **Length:** 25 cm
- **Width:** 20 cm
- **Water Height:** 20 cm

Total volume:

$$25 \times 20 \times 20 = 10{,}000 \text{ cm}^3 = 10 \text{ L}$$

Grow Bed Dimensions

- **Length:** 50 cm
- **Width:** 50 cm

Surface area:

$$0.5 \times 0.5 = 0.25\ \mathrm{m}^2$$

Selection of a prototype

The prototype is to design and construct a small-scale smart farm prototype focused on aquaculture, greenhouse agriculture, or hybrid aquaponics. The prototype must include a closed-loop feedback control system that monitors at least three environmental parameters, with only one permitted to be temperature, pH, or salinity, using calibrated sensors appropriate to the selected system. The 3D **design is shown in Figure 35.**

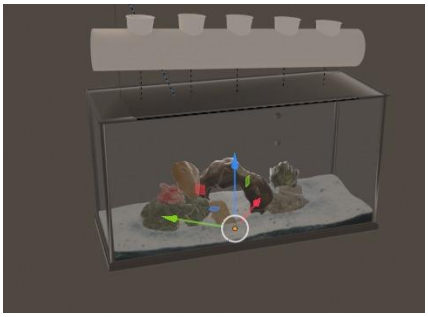


Figure 35 shows the prototype 3D

Mechanism

The solution is hybrid aquaponics, which can be implemented using many different systems; the most valuable is the Nutrient Film Technique (NFT) system, **as shown in Figure 36**, on which the prototype method relies. This is the explanation of its mechanism. It's composed of three main parts: a fish tank containing an air pump and a biofilter tank, and a growing pipe for planting.

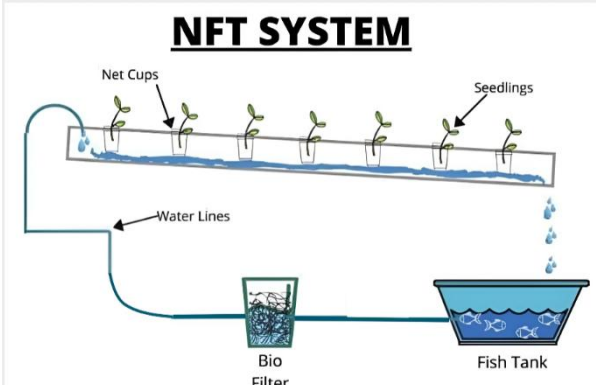


Figure 36 NFT system

Fish produce ammonia through waste and gills. Then the water that is rich in ammonia, which is toxic for fish at high levels, in addition to being useless for plants, is pumped from the fish tank to the bio filter containing beneficial bacteria, which plays a pivotal role in this system due to dealing with the incoming water and converting ammonia to the useful nutrients for plants through processes called the Nitrification.

The ammonia is in equilibrium with ammonium in water.

$NH^3 + H_2O \rightleftharpoons NH^{4+} + OH^-$

Beneficial bacteria called Nitrosomonas convert ammonia into nitrites. Nitrite is still harmful to fish but slightly less toxic than ammonia.

$$NH_3 + 1.5O_2 \rightarrow NO_2^- + H^+ + H_2O$$

Then, Nitrobacter bacteria convert nitrites into nitrates. Nitrates are safe for fish and perfectly usable for plants.

$NO^{2-} + 0.5O_2 \rightarrow NO^{3-}$

Then the water passes through the pipe, and plants can absorb the nutrients.

There is a smart closed loop

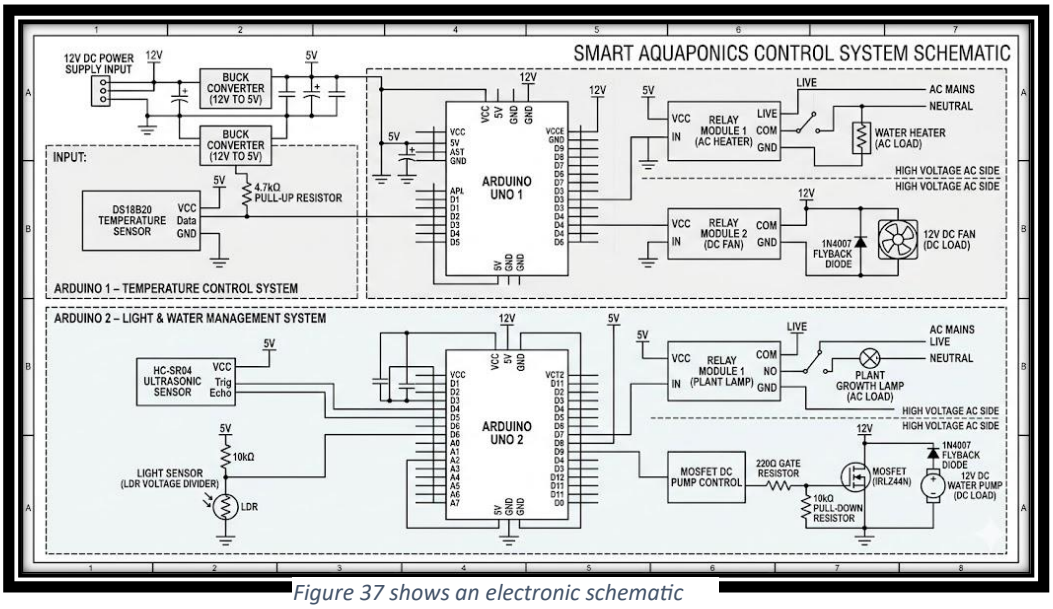


Figure 37 shows an electronic schematic

feedback system that monitors three parameters: water level by an ultrasonic sensor, temperature by a DS18B20 sensor, and light intensity by a Light Intensity Module (GY-302 BH1750). An Arduino will take these sensors' readings and turn on the actuators, that have capable of regulating variations in those three parameters. Actuators must operate in response to sensor feedback through the control algorithm. This prototype contains four actuators: a heater and dc fan for controlling the water temperature, a plant growth lamp for providing the ideal growth light, and a water pump to stabilize the water level. These prosses done by two microcontrollers (Arduino Uno). The process **is shown in Figure 37**.

Materials:

Electronic materials: Ultrasonic sensor, Light intensity sensor, temperature sensor, two water pumps, a DC fan, a water heater, a growing light, Arduino Uno, four relays, Adaptotr, lcd screen, air pump, buzzer, bread board, and wires.

Recycled materials: acrylic for greenhouse gases, open plastic pipe from the top for net cup, Water hose for water movement, fish tank, bio filter tank.

Living organisms: lettuce, tilapia, and beneficial bacteria.

Chapter 3: Constructing and Testing a Prototype

Abstract

Egypt faces several major challenges, including Population growth and its consequences, and addressing and reducing pollution that fouls our air, water, and soil. Egypt suffers from increasing food demand due to population growth, which places pressure on agricultural systems that rely on limited freshwater resources and unsustainable farming practices. The project aims to solve these problems by creating hybrid aquaponics. The prototype features a 0.5 m^2 greenhouse integrated with a 10 L aquaculture tank, creating a closed-loop nutrient cycle. This project will be portable and testable under different conditions, so the system was specifically engineered for the high-humidity environment (avg. 70%) of Borg El Arab City, supporting Egypt's goal of expanding its industrial and agricultural base, furthermore, as a part of the Alexandria government, which houses 5.5 million of resident which addresses the Egyptian Grand Challenge of dealing with population growth and its consequences. The project aims to measure three parameters (Temperature, light intensity, and water level), and they are treated by four actuators (water pump, LED, heater, and DC fan). The project includes a preloaded database with optimized environmental profiles for plant species (Lettuce and Basil) and aquatic organisms (Tilapia and Salmon). The prototype has a positive result like The prototype recorded an average temperature of **20.34°C**, with a minimum of **18.31°C** and a maximum of **36.75°C**, showing generally stable conditions with occasional increases. The light intensity (lux) had a mean of **827.39 lux**, ranging from a minimum of **31 lux** to a maximum of **5884 lux**, indicating wide variation in lighting levels. Additionally, the ultrasonic sensor measured distance with an average of **17.98 cm**, a minimum of **5.20 cm**, and a maximum of **39.98 cm**, reflecting changes in water level during system operation and some negative results such as fish death and some electronic damage.

Introduction

Egypt is currently facing several grand challenges affecting the food demand and freshwater resources, as well as unsustainable farming practices. One of them is population growth, **which is shown in Figure 38**, and its consequences. An increase in population growth places a heavy load on the Egyptian economy, and it influences the food demand. Also, addressing and reducing pollution fouling our air, water, and soil is one of these challenges. Reports say that 38 million Egyptians drink water polluted by sewage, leading to the spread of diseases such as cholera and hepatitis, which negatively affect the limited freshwater resources and unsustainable farming practices.

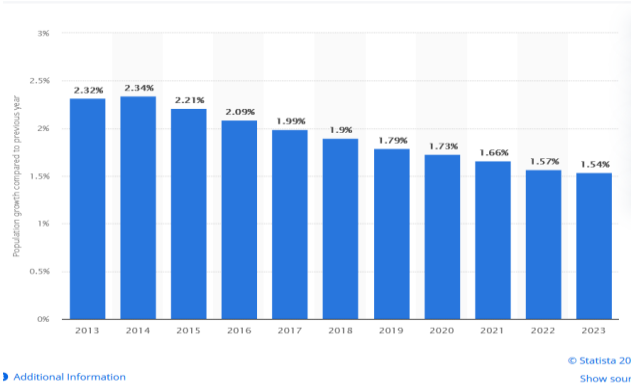


Figure 38 shows the population growth percentage, Statista. (2024, September 27).

Therefore, in this semester, we were asked to design and implement an automated closed-loop control system to monitor and regulate essential environmental variables in aquaponics. So, our prototype focused on a specific problem, food scarcity, and traditional irrigation.

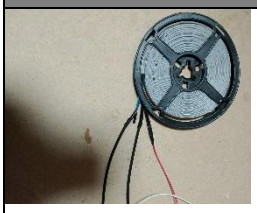
In researching the Problem to Be Solved, we analyzed prior solutions like Upward Farms and Aquaponics Farm Al-Bustan. These systems demonstrate a major strength in water conservation, using recycling to save up to 90% more water than traditional farming. However, a significant weakness is the high operational cost of artificially recreating sunlight and airflow. Upward Farms had to pay to artificially recreate all those things 24 hours a day, 7 days a week.

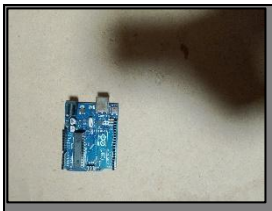
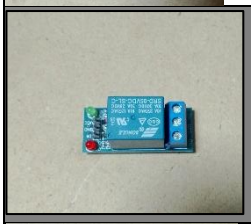
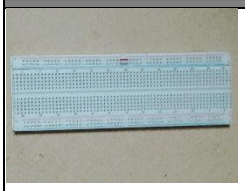
To meet the demands of a growing population and rising water scarcity, agriculture must transition from traditional irrigation to more efficient, targeted methods. Our team chose a hybrid aquaponics solution that utilizes a symbiotic cycle: fish waste provides nutrients for plants, while plants naturally purify the water for the fish. We chose this specific design because it offers a portable, cost-effective, and closed-loop ecosystem that minimizes waste. This harmonious relationship between aquatic life and vegetation fosters a closed-loop ecosystem, significantly curbing water wastage and elevating the system's overall sustainability.

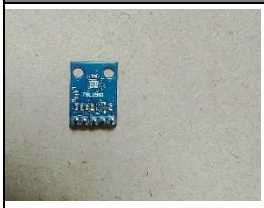
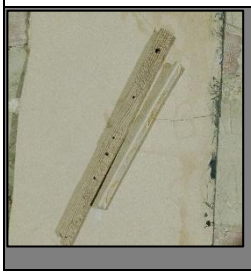
To ensure success, our prototype had to meet strict Design Requirements: a 0.5 m^2 footprint, a 10 L aquaculture tank, and an automated system to manage three parameters (temperature, light intensity, and water level). These are regulated via setpoints in an Arduino-based controller that triggers a water pump, LED array, heater, and DC fan. Additionally, the system features a preloaded database for Lettuce, Basil, Tilapia, and Salmon to automate environmental profiles. By utilizing recycled wood, plastic, and glass, we ensured the prototype is both sustainable and accessible. To achieve this functional design, specific Materials and Methods were employed to construct and rigorously test the system's performance.

Materials & methods

Materials

Picture	Name	Quantity	Description	Price
	Heater	1	Raising the water temperature of the fish tank	475 EGP
	LED	1	The responsible part for the plant's photosynthesis	147 EGP
	DC fan	1	Reducing the fish water temperature	recycled
	Servo motor	1	Control of the feeding system	75 EGP

	Arduino	1	The brain of the circuit	Recycled
	LCD	1	Display sensors readings	Recycled
	Relay	3	control on the circuit by an independent low-power signal	45EGP
	Breadboard	2	Connection of the circuit	Recycled
	Power supply	1	Circuit Power source	290 EGP
	Temperature sensor	1	Read the water temperature	Recycled
	Water tube	3	Moving of the cycle water	Recycled
	Fish tank	1	The fish's home	Recycled

	PVC pipe	1	Carry the plants and the body of the greenhouse	Recycled
	Light sensor	1	Read the light intensity	190 EGP
	Water bottle	1	Work as a sump	Recycled
	Water pump	2	Push the water between the greenhouse and the fish tank	290 EGP
	Pieces of wood	4	Carry the prototype and cover the greenhouse	Recycled

Methods

To build the prototype, you should follow these steps.

1. We used a recycled wooden box to carry the prototype as a frame; the frame has a height of 130 cm, a length of 70 cm, and a width of 50 cm.

Figure 39 shows cutting the greenhouse

2. We cut pieces of wood for a greenhouse, which has an isosceles triangle with each side measuring 40 cm, 15 cm height, and 50 cm width, **as shown in Figure 39**, fixed on the wooden box, also covered with a plastic cover, and built a greenhouse gate.

Figure 40 while painting the frame

3. We painted the wooden box a brown color and the greenhouse a white color, **as shown in Figure 40**.

4. Cut a recycled PVC pipe and make three equal holes on the top of it to put the plants in these holes.

5. We washed a recycled fish tank, **as shown in Figure 41,** and put in beneficial bacteria and Chlorine remover to make the water ready to put the fish in it, and added the heater to the fish tank. Also, fixing the servo motor and the DC fan above the fish tank.

Figure 11 while washing the fish tank

6. Connect the Arduino with sensors: a light sensor, a temperature sensor, and a water level sensor, and actuators: a DC fan, pumps, an LED, a heater, and a servo motor, **as shown in Figure 42**.

7. Cutting the water tube into three parts, one reaches from the fish tank to the PVC pipe, the second one from the PVC to the bottle of water, and the final one from the bottle to the fish tank.

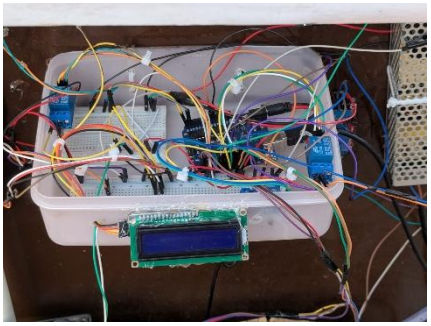


Figure 42 shows the circuit connection.

8. Put two pumps, one from the fish tank into the greenhouse, and the second one from the bottle to the fish tank.
9. Make a biofilter with cotton, gravel, and clay, and put it in a small water bottle, **as shown in Figure 43.**

Figure 43shows the biofilter.

10. Sensor's calibration**. As shown in these graphs 44, 45**

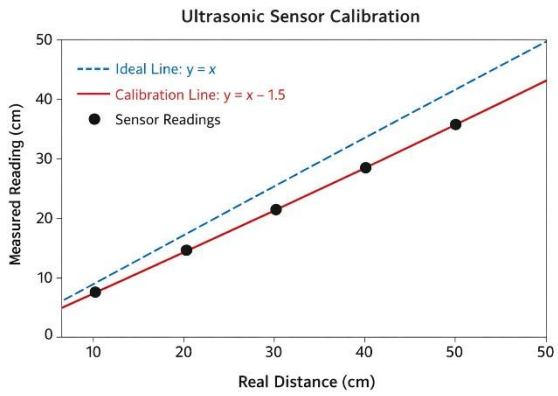


Figure 44 ultrasonic sensor calibration

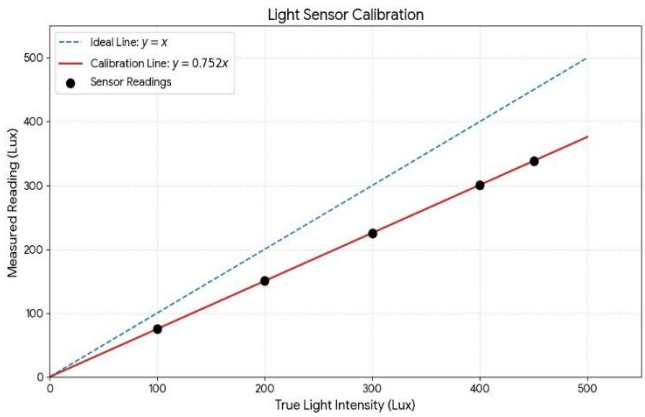


Figure 45 light sensor calibration

Safety precautions taken during construction

Our team's interest in saving our salves while building the prototype, so we follow all protection processes. In our prototype, we used MDF glue to fix the materials, and this glue is toxic to our health, so we wore a mask to avoid spraying and a coat to avoid damaging our clothes. We also used nails to fix the wood in the prototype, so we wore gloves to avoid any injury to our hands. While changing the water of the fish tank, we wore gloves to avoid the ammonia of the water, and masks to avoid the bad smells. We maintain a clean and clutter-free workspace to prevent accidents. We make sure that there is good lighting to work properly.

Design requirements

The chosen solution was an aquaponic system. It is an automated system with a closed-loop system. Thus, before designing the prototype, setting requirements is needed to define its success and the results intended to be achieved. Most of the prototype's body was built from recycled materials, such as PVC pipe, a fish tank, a wood frame, and some recycled electronics, like an Arduino, an LCD screen, wires, DC fan. The fish tank is 10 L, and the greenhouse is less than .05 m^2, which achieves the design requirements. The system treats three parameters (water level, temperature, and light intensity), and contains four actuators, a DC fan, a heater, an LED, and pumps. Each parameter has been calibrated with a real device. The system has the order of (IF, then), exactly in the pump. If the water level in the fish tank decreases by a certain distance, the pump will turn on, and we follow all the design requirements to achieve all requirements.

Test plan

After constructing the prototype, it should be tested to ensure its efficiency.

First, the body of the prototype was tested by forcing it with a high weight to ensure its stability.

Before any electronic connection, the sensor's calibration was done by scientific procedures.

An electric circuit test was done through three stages: Firstly, ensure all three sensors are connected and take down their readings with the LCD screen.

The second step was testing all actuators with an if statement and a while loop and writing down the time taken to apply system recovery.

The third step was connecting all sensors and actuators with each other and ensuring they work in concert to achieve maximum efficiency.

The upper pipe was tested to determine if it dropped water.

After these tests, the prototype is professional and precise.

Data Collection of the tools

The ruler, hand saw, hammer, marker, Scissors, and sandpaper are used in our prototype. The ruler was used to measure the dimensions of the greenhouse and the frame. The hand saw was used to cut the wood and PVC pipe. The hammer was used to push the nails while fixing the greenhouse or the circuit. The marker was used to draw the dimensions of the prototype, like the framed dimensions or the water tube dimensions. The scissors were used to cut the wires of the circuit and the tubes. Finally, the sandpaper was used to make the hardwood softer.

The ruler has errors that represent 30 cm± 0.5 mm. The Hammer has errors in the range of 48 to 58 MPa with an error of 15.32%. The scissors have errors that represent 40 to 54 MP, which represent 10%.

Date Collection in the test plan

Results

Negative Results

Dead fish

Before completing the prototype, fish were introduced into the tank under basic conditions, including feeding, a heater, and an air pump. However, after three days, some fish began to die.

After observation and analysis, it was concluded that the primary cause was insufficient oxygen dissolved in the water. In addition, the absence of an efficient filtration system led to poor water quality.

This resulted in the accumulation of waste materials, which likely increased ammonia levels and further stressed the fish. It was also observed that the water required filtration at least twice per week to maintain suitable living conditions.

These findings highlighted the importance of proper aeration and continuous water filtration in maintaining fish health within the aquaponic system.

Lettuce Wilting

Before starting the prototype, lettuce plants without fully developed roots were purchased and placed in water to stimulate root growth. However, after three days, signs of wilting were observed in some plants.

This was mainly due to excessive water exposure and the lack of proper oxygen availability around the base of the plant. Without sufficient oxygen, root development was limited, leading to stress and reduced water and nutrient uptake. As a result, the leaves began to wilt.

These observations indicated that proper aeration and controlled water conditions are essential for healthy root formation and plant growth. Consequently, new lettuce plants were later introduced under improved conditions.

Damage to some electronics

Damage to some electronic components occurred during circuit testing, including an ultrasonic sensor, a power supply unit, and an LCD screen. The causes of failure were

excessive input voltage applied to the ultrasonic sensor, damaged resistors within the power supply circuit, and physical damage where the LCD wires were cut.

Positive result

Light sensor reading through 6 hours:

This sensor was installed in the greenhouse to be exposed to light, such as sunlight. It helps the system detect the light intensity. It's observed that the deviation of the light is frequent. **As shown in Graph 46.**

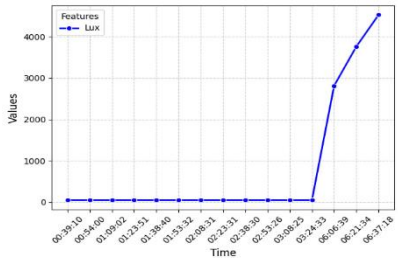


Figure 46 The readings of the light

ultrasonic sensor reading through 6 hours:

Ultrasonic is responsible for detecting the distance between the top border of the fish tank and the water level. This parameter wasn't affected continuously, that's because decreasing water depends on the sunlight. **As shown in Graph 47**.

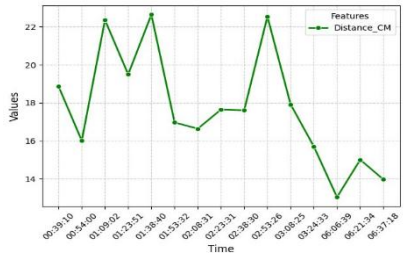


Figure 47 ultrasonic sensor readings

Temperature sensor reading through 6 hours:

It's important to know any deviation in the water temperature to preserve the fish. **As shown in Graph 48**.

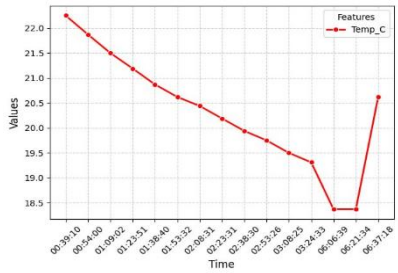


Figure 48: Temperature sensor readings

Power consumption:

Table 6 shows the Power consumption analysis of the prototype components, showing the voltage, current, and power requirements for each actuator. The results indicate that the total power consumption reached 279.5 W, with the fish tank heater and LED strip being the highest energy-consuming components, reflecting their significant role in maintaining system operation.

Component	Voltage (V)	Current (A)	Power (W)	Energy per Second (J/s)
Water Pump 1	12	0.5	6	6
Water Pump 2	12	0.5	6	6
LED Strip (4m)	12	4.8	57.6	57.6
Cooling Fan	12	0.2	2.4	2.4
Fish Tank Heater	220	0.9	200	50
Servo Motor (Blue)	5	0.5	2.5	2.5
Air Pump	220	0.4	5	5
Fish tank lamp	220	0.45	10	
Total			279.5	129.5

Table 6 shows the power consumption

Time response:

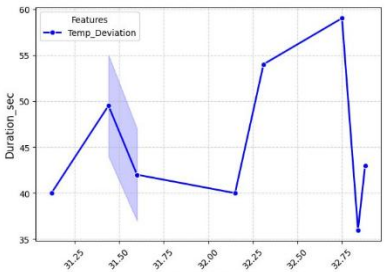


Figure 49 shows the time response for temperature readings

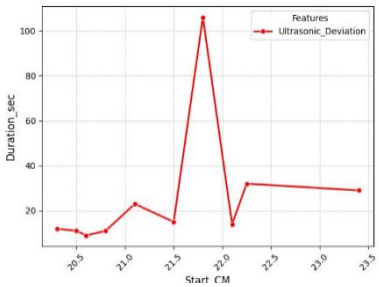


Figure 50 shows the time response for temperature readings

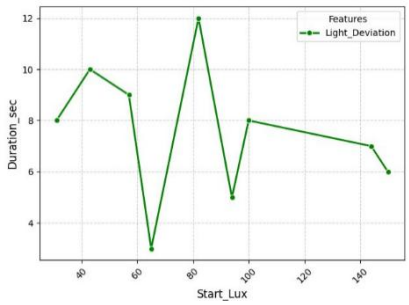


Figure 51: shows the time response for temperature readings

Chapter 4 Evaluation, Reflection, Recommendations, and Discussion

Analysis and discussion

Nitrogen cycle

Nitrogen is an essential element for all living organisms because it is a key component of DNA, RNA, amino acids, proteins, and other cellular structures. In aquaponic systems, the primary source of nitrogen is fish feed. After digestion, fish excrete nitrogen mainly as ammonia (about 90%).

In water, ammonia exists in two forms: un-ionized ammonia (NH_3) and ionized ammonium (NH_4^+). The total of both forms is called Total Ammonia Nitrogen (TAN). The balance between NH_3 and NH_4^+ depends on environmental conditions such as pH and temperature. Higher pH and temperature increase the proportion of NH_3, a highly toxic compound to fish.

Ammonia and nitrite are dangerous because they can damage fish gills and interfere with oxygen transport in the blood. Therefore, these compounds must be quickly converted into less harmful substances.

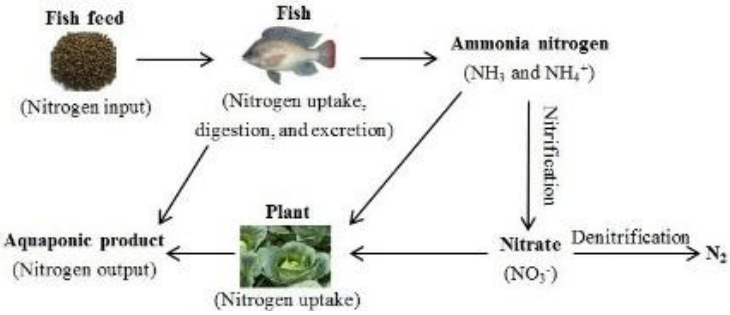


Figure 52 shows the Nitrogen cycle.

The nitrogen cycle in aquaponic systems depends on beneficial bacteria. Ammonia-oxidizing bacteria (AOB), such as *Nitrosomonas*, convert ammonium into nitrite (NO_2^-). Then, nitrite-oxidizing bacteria (NOB), such as *Nitrobacter*, convert nitrite into nitrate (NO_3^-). This process is known as nitrification, and it is **shown in Figure 52.**

Nitrate is much less toxic and is absorbed by plants as a nutrient. Plants use nitrate to produce organic nitrogen compounds such as amino acids and proteins. This process improves Nitrogen Use Efficiency (NUE), which refers to how effectively nitrogen is converted into plant biomass.

The main chemical reactions involved are:

Ammonia equilibrium in water:

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

Ammonia oxidation (first nitrification step):

$$NH_4^+ + 1.5O_2 \rightarrow NO_2^- + 2H^+ + H_2O$$

Nitrite oxidation (second nitrification step):

$$NO_2^- + 0.5O_2 \rightarrow NO_3^-$$

Plant uptake (assimilation):

$$NO_3^- \rightarrow \textbf{Organic Nitrogen (amino acids, proteins)}$$

The microcontroller connections

The microcontroller is called the brain of the embedded system, acting as the communication interface between sensors and actuators. To begin with, sensors collect information regarding the physical parameters from the real-world environment. Further, the collected information is processed by the microcontroller, and decisions related to the actions taken by the actuators are made based on that interpretation. **As shown in Figure 53**, the actuators are motor-based devices. An actuator is a device in the control system that assists the control process in executing its activities or actions. These devices use electric energy and convert it to mechanical movement. Four distinct stages include:

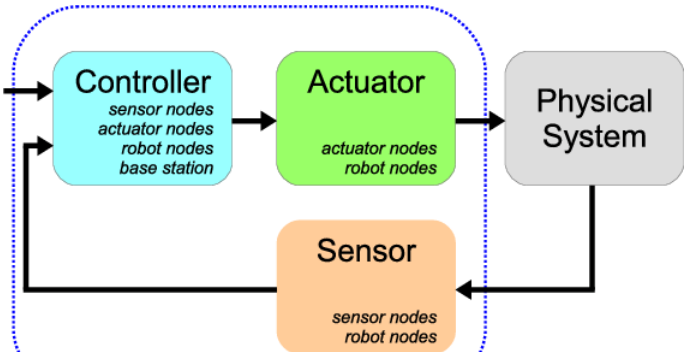


Figure 53 shows the microcontroller connections

Collection of Data through Sensors: Signals in sensors occur, and they are either analog or digital for the microcontroller to use. Analog Sensors: Produce signals in terms of voltage level, which require converting to digital through the ADC process, while Digital signals represent only two states (HIGH or LOW).

$$Vin = \frac{ADC\ value}{1023} \times Vref$$

Processing of the Signals from Sensors: This involves the microcontroller performing filtration and calibration on the signals, as well as algorithms. For example, it may compare the sensor value to a specific condition.

$$Output = \{ ON \ if \ Sensor \ Value > Threshold$$
$$OFF \ otherwise \}$$

In addition, Information collected could be sent to a database using communication techniques such as wired and wireless communications. Finally, the Arduino sends output signals based on the processed data. These outputs can control devices such as LEDs, motors, or buzzers. Additionally, Arduino can transmit data to other devices or systems using communication methods such as serial communication, I2C, or SPI.

Ultrasonic Sensor

Mechanism:

Ultrasonic sensors work as eyes in living organisms. But its mechanisms of work are different. It works by sound waves at 40 kHz, which is never detected by human ears, as it is out of the human ear's allowed sound range. This sensor emits these waves to go and after hitting somebody, this sound reflects to the sensor again, **as shown in Figure 54**, and by these variables, this sensor can determine the distance according to this equation

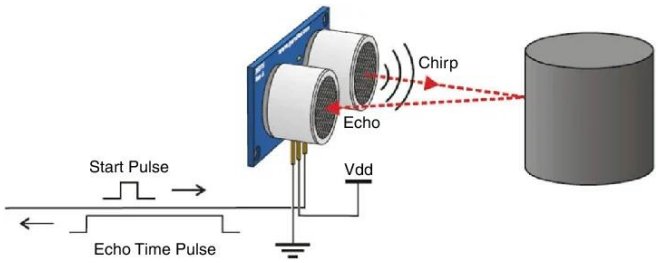


Figure 54 shows the mechanisms of the ultrasonic.

d = (v × t) / 2.

Where d is the distance to the water surface in meters, v is the speed of sound in air (approximately 343 m/s), and t is the total round-trip time in seconds. The division by 2 accounts for the fact that the sound travels to the surface and back.

Role: This sensor is used to detect dropped below a defined minimum threshold in the water level of the fish tank, and it is connected to a water pump as an actuator, which pumps and increases the water level of the tank.

DS18B20 Temperature Sensor

Mechanism

The DS18B20 is a digital temperature sensor that measures water temperature with a resolution of up to 0.0625°C. The DS18B20 sensor converts ambient thermal energy into a digital value by measuring the frequency shift of an internal temperature-dependent oscillator. This digital data is transmitted to a microcontroller via the 1-Wire protocol, which allows bidirectional communication over a single signal line labeled IO. The pull-up resistor (R7) It is essential to this process, as it maintains the signal line at a "High" voltage state so the sensor can pulse the line "Low" to send data bits, **as shown in Figure 55**. At its core, the sensor contains a temperature-sensitive resistor whose Electrical resistance changes predictably with temperature. And it has a built-in system that converts analog to digital signals. The temperature-to-digital conversion follows this relationship for its 12-bit resolution mode: T = raw value × 0.0625 , Where raw value is the integer output from the sensor, and T is the temperature = 24°C exactly. Role: This sensor controls 2 actuators and 2 functions (heating & cooling).

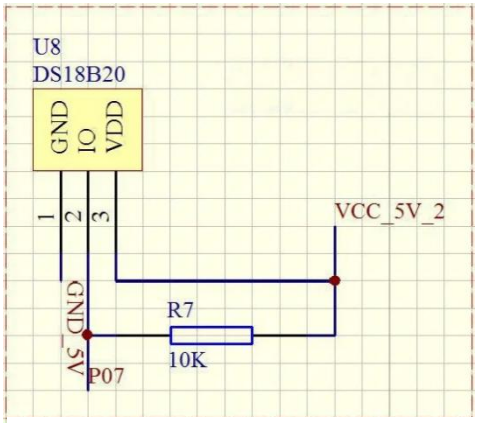


Figure 55 shows the mechanisms of the DS18B20 Temperature Sensor

If the sensor reports a temperature below 24°C, the Arduino activates the water heater to warm the tank. If the temperature exceeds 30°C, the heater is switched off, and the DC fan is activated to cool the water through evaporation and airflow across the surface.

TSL2561 Light Sensor

Mechanism:

The TSL2561 is a light-to-digital converter that measures in lux, the standard unit for visible light intensity.

It contains two photodiodes with different spectral sensitivities.

The first photodiode, called Channel 0, responds to the full visible and infrared spectrum.

The second photodiode, called Channel 1, responds primarily to infrared light only.

And the plant just takes the benefit or does its vital process according to the visible light, so the infrared we should subtract from the total light. And the sensor does that according to the following equation: Lux = **C0 × CH0 − C1 × CH1.**

Where CH0 and CH1 are the raw counts from the two photodiode channels, and C0 and C1 are calibration coefficients. This subtraction approach is what makes the TSL2561 significantly more accurate than a simple single-photodiode sensor.

Role: And this sensor is connected to the LED light, which switches it on when the lux below down about safe zone or the minimum lux required for the plant can do their vital processes.

<u>Efficiency</u>

First step: Evaluating the reading of the sensors and comparing it to reference sensors, such as the sensors in the laboratory. And calculate the error by Mean Squared Error (MSE) or by Mean Absolute Error (MAE).

MSE: $(1/n) \times \Sigma (y_i - \hat{y}_i)^2$

MAE: $(1/n) \times \Sigma |y_i - \hat{y}_i|$

Second step:

Normalization of the error by dividing the error by the maximum reading of the sensor, because the error of the sensor is in its range from 0 to 50, as a temperature sensor, it is impossible to compare it with a sensor whose range is from 0 to 40000.

Third step: Calculate the efficiency of each sensor by (100% - (normalization error)%).

Calculation of the whole system by taking the average efficiency of all sensors.

Conclusion

The project aimed to address population growth and food scarcity by developing a closed-loop automated aquaponic system that integrates aquaculture and agriculture. The system treats three parameters (water level, temperature, and light intensity), and contains four actuators, a DC fan, a heater, an LED, and pumps. Each parameter has been calibrated with a real device. The system has the order of (IF, then), exactly in the pump.

For the bodies, the fish tank is 10 L, and the greenhouse is less than .05 m^2, which achieves the design requirements.

During simulated testing, the prototype recorded an average temperature of **20.34°C**, with a minimum of **18.31°C** and a maximum of **36.75°C**, showing generally stable conditions with occasional increases. The light intensity (lux) had a mean of **827.39 lux**, ranging from a minimum of **31 lux** to a maximum of **5884 lux**, indicating wide variation in lighting levels. Additionally, the ultrasonic sensor measured distance with an average of **17.98 cm**, a minimum of **5.20 cm**, and a maximum of **39.98 cm**, reflecting changes in water level during system operation

Overall, the prototype demonstrated the ability to monitor environmental parameters and respond automatically to maintain stable conditions suitable for aquaponic plant growth.

Recommendations

Predictive control systems

Predictive control systems are recommended for this project to anticipate deviations and trends in parameters like the fish tank's water level, water temperature, and the plant's light intensity before they occur. Predictive systems use the current process measurements, the current dynamic state of the process, and the process variable targets and limits to calculate future changes in the dependent variables. These changes are calculated to keep the dependent variables close to their targets while honoring constraints on both independent and dependent variables. Although a simplified rule-based approach is used instead of a full mathematical MPC model, the system maintains the core concept of predictive decision-making to enhance stability, efficiency, and resources**, as shown in Figure 56.**

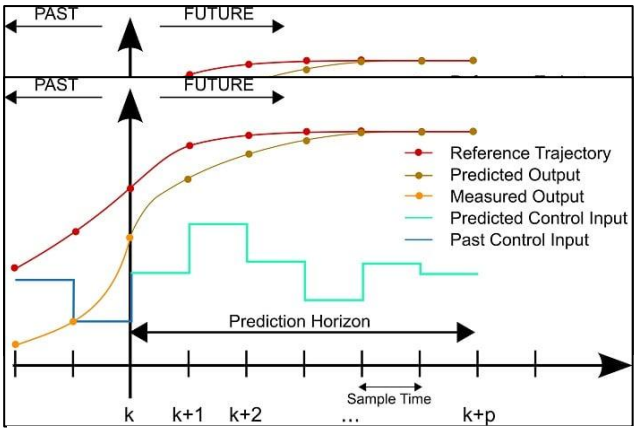


Figure 56 shows the importance of the predictive system.

A Water generator

A water generator is recommended to be implemented in this aquaponics system. This component relies on gravity and water flows from the upper plant pipe to the lower fish tank. This generator converts the mechanical energy into electric energy by letting water enter it and turn the turbine, and then it goes out. Some terms are needed to complete this process at full face: minimum height 2 meters, the pipe width should be doubled than ours. This addition will make this project more sustainable and decrease power consumption; as a result, it provides electricity for the microcontroller to work continuously. The generator **is shown in Figure 57**.

Figure 57 shows the water generator

Real-life application

We chose Borg El Arab City in Alexandria government to be our real-life location, **as shown in Figures 58 and 59**, due to its environmental and demographic characteristics that contrast with our school's location. Unlike the arid desert climate of October Gardens, Borg El Arab features a coastal environment with an average humidity of 70% and a moderate, stable climate year-round. This selection allows us to address the challenge of increasing the industrial and agricultural base of Egypt. Furthermore, as a part of the Alexandria government, which houses 5.5 million of resident which addresses the Egyptian Grand Challenge of dealing with population growth and its consequences. The city provides a broad land availability for scalable agriculture, serving as a strategic hub to increase Egypt's agricultural base.

Figure 58 shows Borg El Arab

Figure 59 shows the beach of Borg El arab

Recommendations for Other Teams

As a capstone team, we will suggest some advice to another future team to improve their capstone. Firstly, the distribution of rules in an organized manner is critical to capstone teamwork. Second, avoid any tension between team members. That will affect your team negatively. Third, while making the prototype, you should choose the materials carefully before starting the prototype to avoid getting into trouble. Fourth, gain the experience from grade 11 or grade 12 to help you in a capstone. That will help you make posters, portfolios, and prototypes perfectly. In the end, avoid doing your tasks lately. This reason is the most important thing in the capstone.

Project's Positive Influence

Despite the many hurdles our team faced while working on our Capstone project, such as our inexperience, we have all grown as individuals and students. As individuals, we adapted to working in a team, became more competent and critical, and realized each member's expertise and shortcomings. While it cemented invaluable ideals and mindsets in us, we would not have learnt otherwise, such as knowing when to defend a task you have done and when to accept others' opinions and improve on it.

As Students, we started to draw connections between our different subjects to achieve our goals in the capstone. For instance, we used both mechanics and physics to calculate different properties of our prototype. Coincidentally, applying what we learnt enabled us to better memorize them, which made our exams and quizzes easier, even if we didn't properly study for them. Thus, our capstone project gave us personal growth.

Los table

LO	**Math LO 1.02**	**PHY 2.03**	**Chem LO 1.01**	**Mech LO 2.01**	**GEO LO 2.01**
Description	Using the type of graphs and the different ways to measure the error in sensors, such as MAE & MSE	DC-Circuits, connections in the circuits, voltage, and power consumption.	Determine the type of error during the calibration and the offset or coefficient	Calculating the mechanical power of the movement of the water in the system	Mechanism and the structure of the filtering of the water to evolve and create (BIO Filter)

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